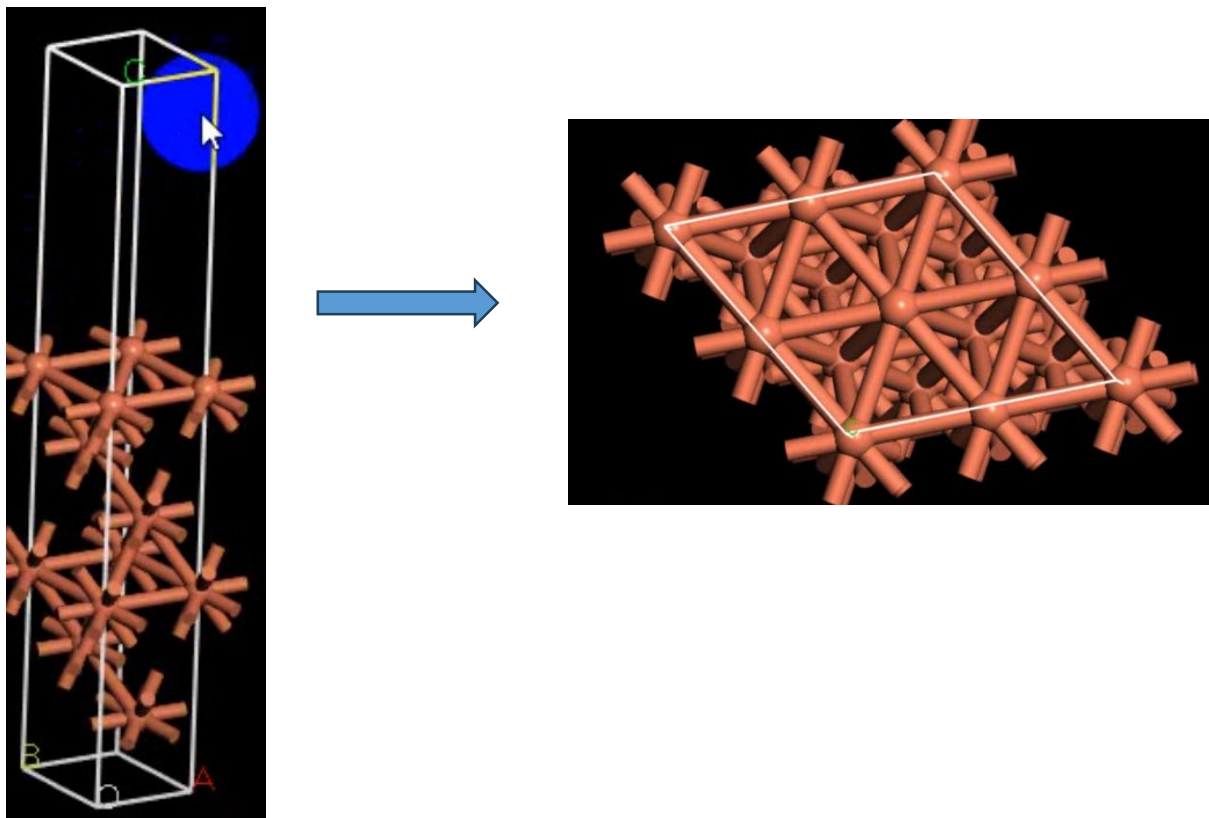


吸附及电子结构后处理分析

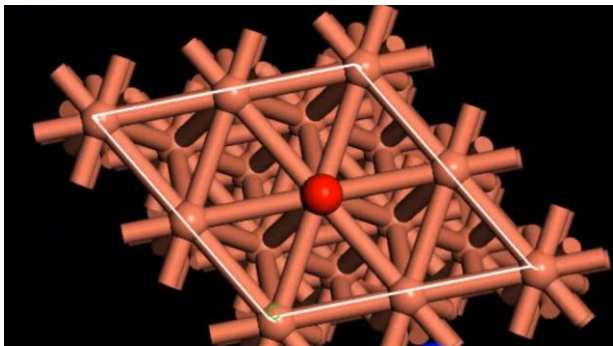
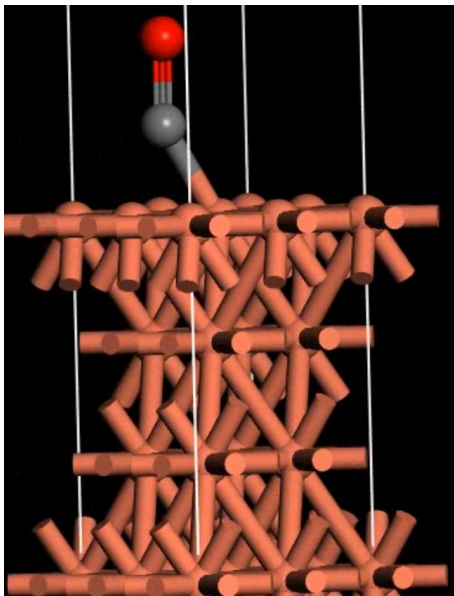
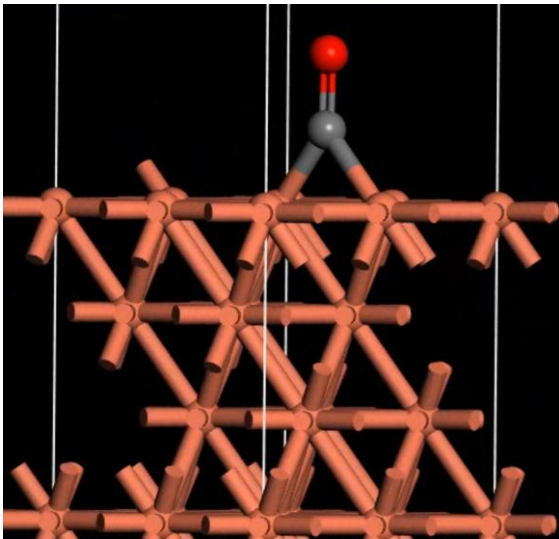
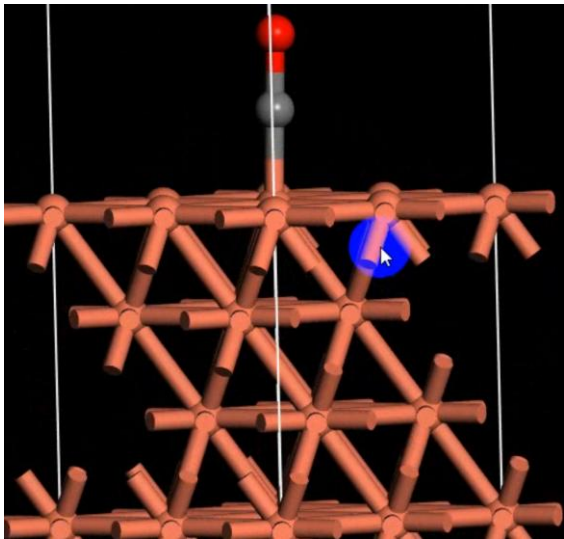
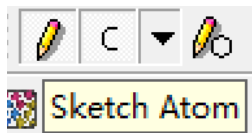
DFT在表面吸附上的简单应用：

以CO吸附在Cu(111)表面为例，判断top, bridge, hollow三种吸附构型哪种更稳定？

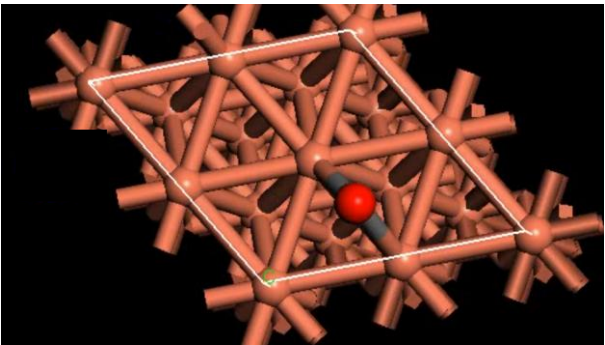
将之前表面优化好的多层Cu(111)表面扩胞为2x2x1,



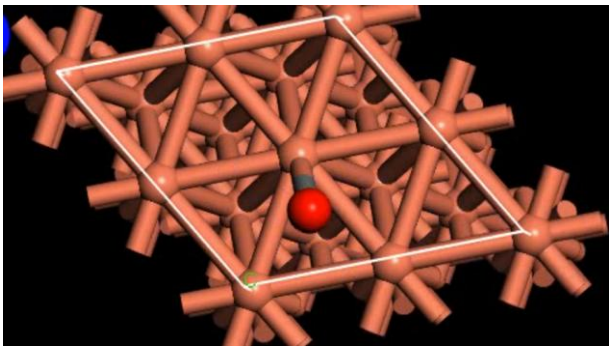
用顶部的**Sketch**工具，添加原子，搭建三种CO吸附构型的结构：top，bridge，hollow



Top

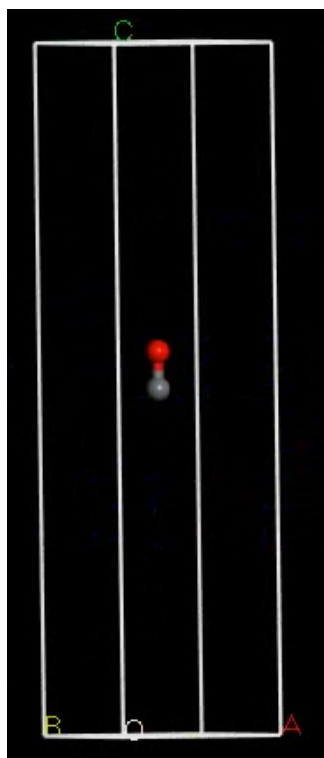


Bridge



Hollow

此处的CO键长还需确定，我们需要得到一个合理的CO分子键长，在VASP中对CO进行结构优化，可以在与之前表面优化同样大小的晶胞中进行，优化好后再修改吸附结构里的C-O键长，将修改后的结构导出，放入VESTA中转化为VASP的输入文件，然后进行结构优化计算（与之前一样，底部几层固定）。根据下面的公式，计算各种吸附构型的吸附能，对比得出最稳定构型。



$$E_{ads} = E_{CO+Cu_surf} - (E_{CO} + E_{Cu_surf})$$

电子结构性质分析

1.电荷密度分布

2.Bader电荷分析

3.差分电荷计算

静态计算/单点能计算NSW=0/1

电荷密度分布

电荷密度分布:需要得到CHGCAR

INCAR设置参数:

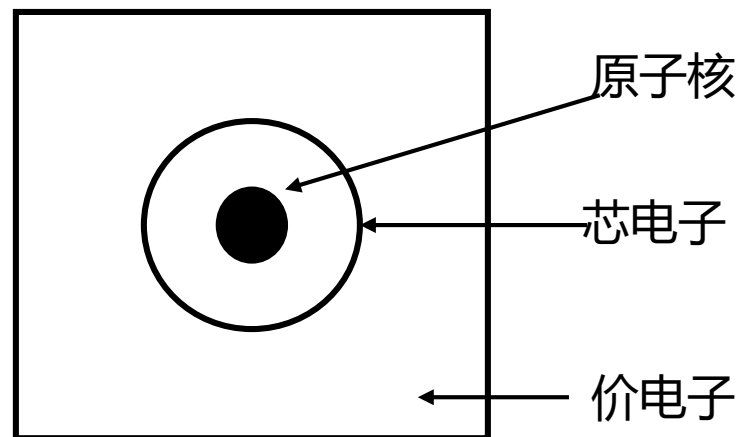
NSW=0

LCHARG=.TRUE.

1.**CHGCAR**是基于赝势计算得到的价电子电荷密度,所以在芯电子的位置不能表现出波函数的节点特征

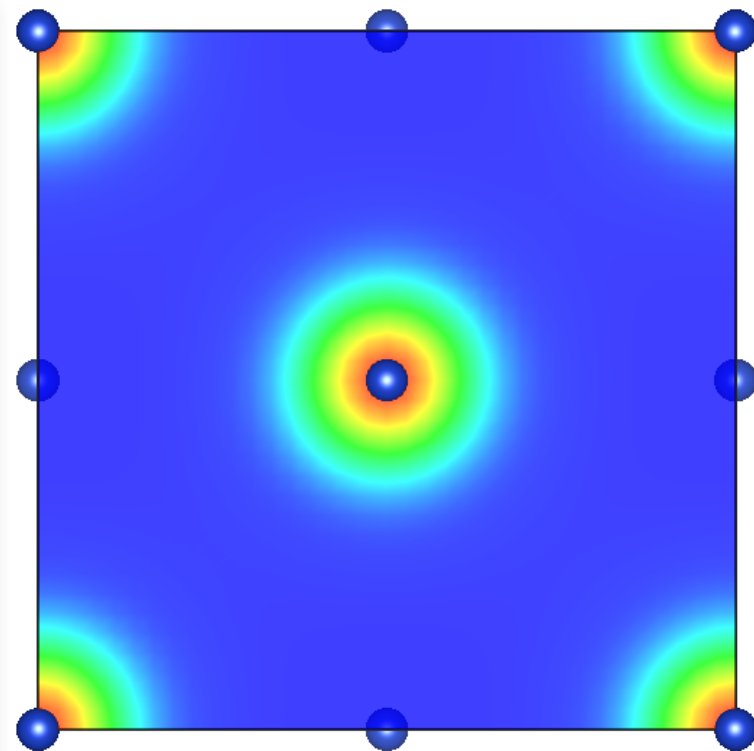
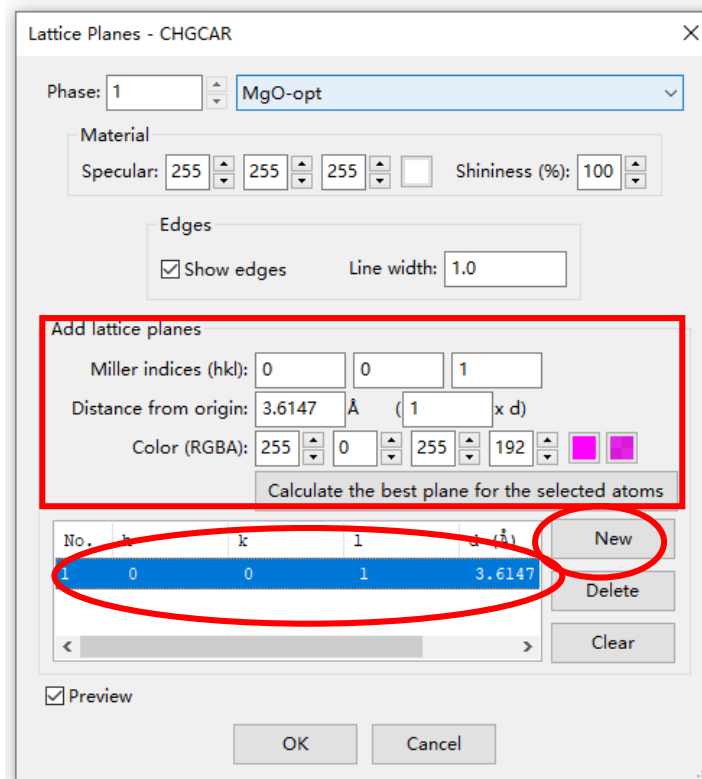
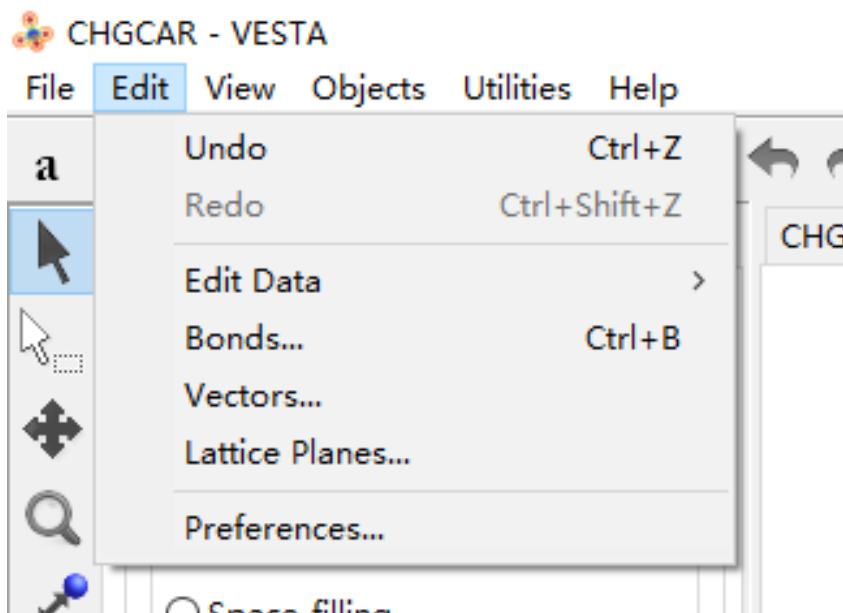
2.如果需要得到芯电子的正确的节点特征,添加**LAECHG=.TRUE.**, 计算结束后输出**AECCAR0, AECCAR1, AECCAR2**, 处理之后得到全电子密度

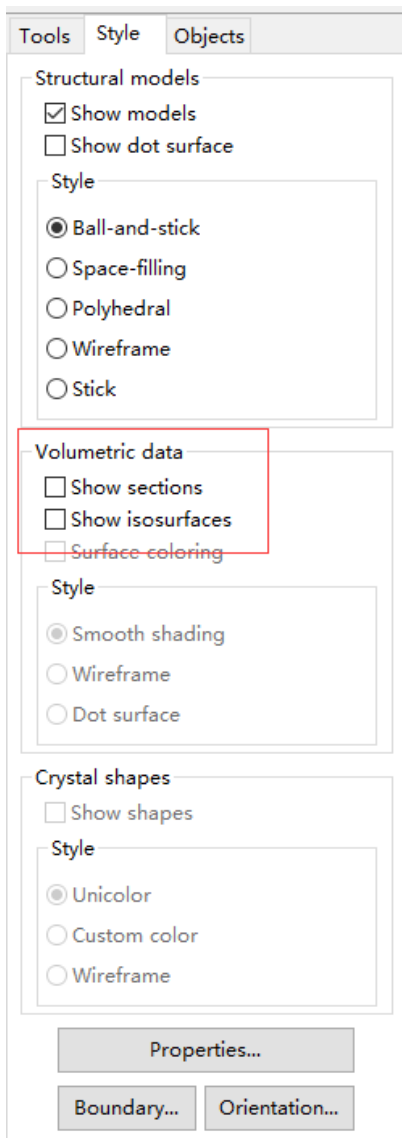
3. **CHGCAR**是包含电子密度信息的格点文件,对于自旋极化体系,计算还包含自旋电子密度。可以用**VESTA, Jmol**等程序打开



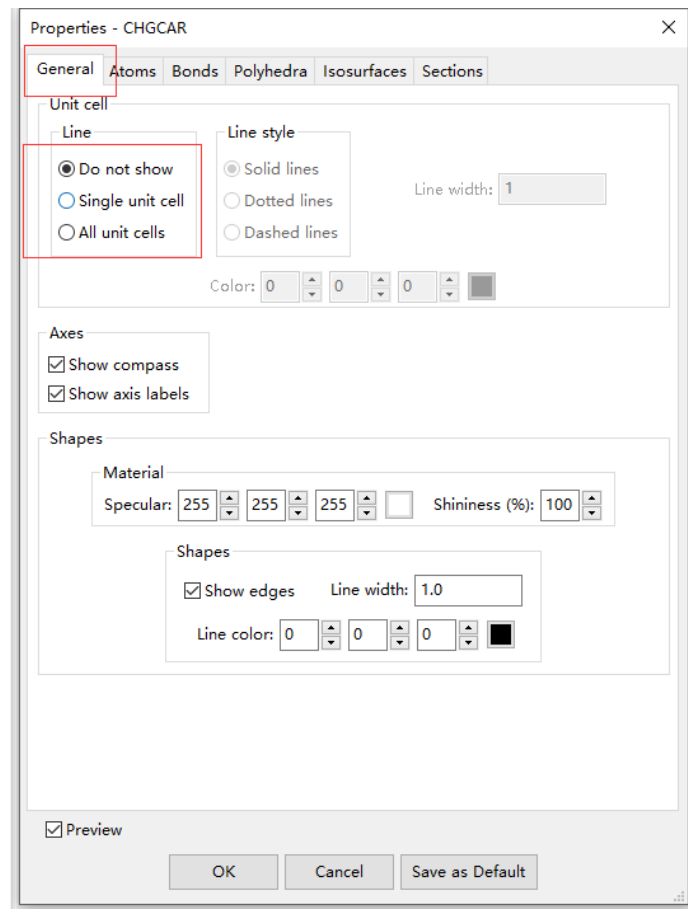
画二维电荷密度截面图 (方法1-原子可以看到)

导入CHGCAR → Edit → lattice plane → New 添加平面(选择米勒指数或者三个原子确定一个平面)

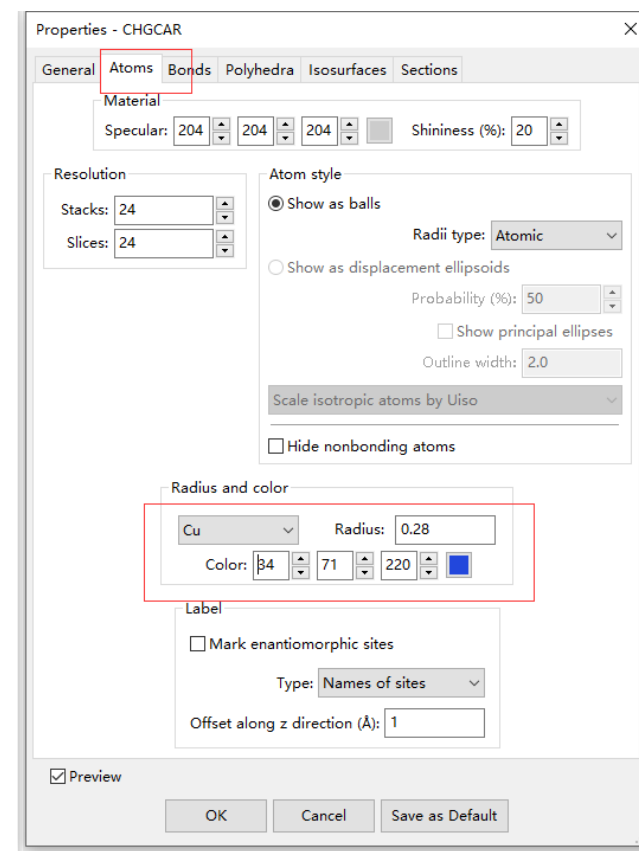




关掉isosurface



去掉晶格边界



改变原子大小，建议把原子变小

方法二：有colorbar

Utilities → 2D Data Display → Slice (选择米勒指数或三个原子，这个方法可以显示colorbar，但是不能显示原子)

colorbar显示颜色到数值的映射

The image shows a sequence of steps in VESTA software to generate a 2D data display slice with a color bar. The main window is titled "2D Data Display - (CHGCAR)".

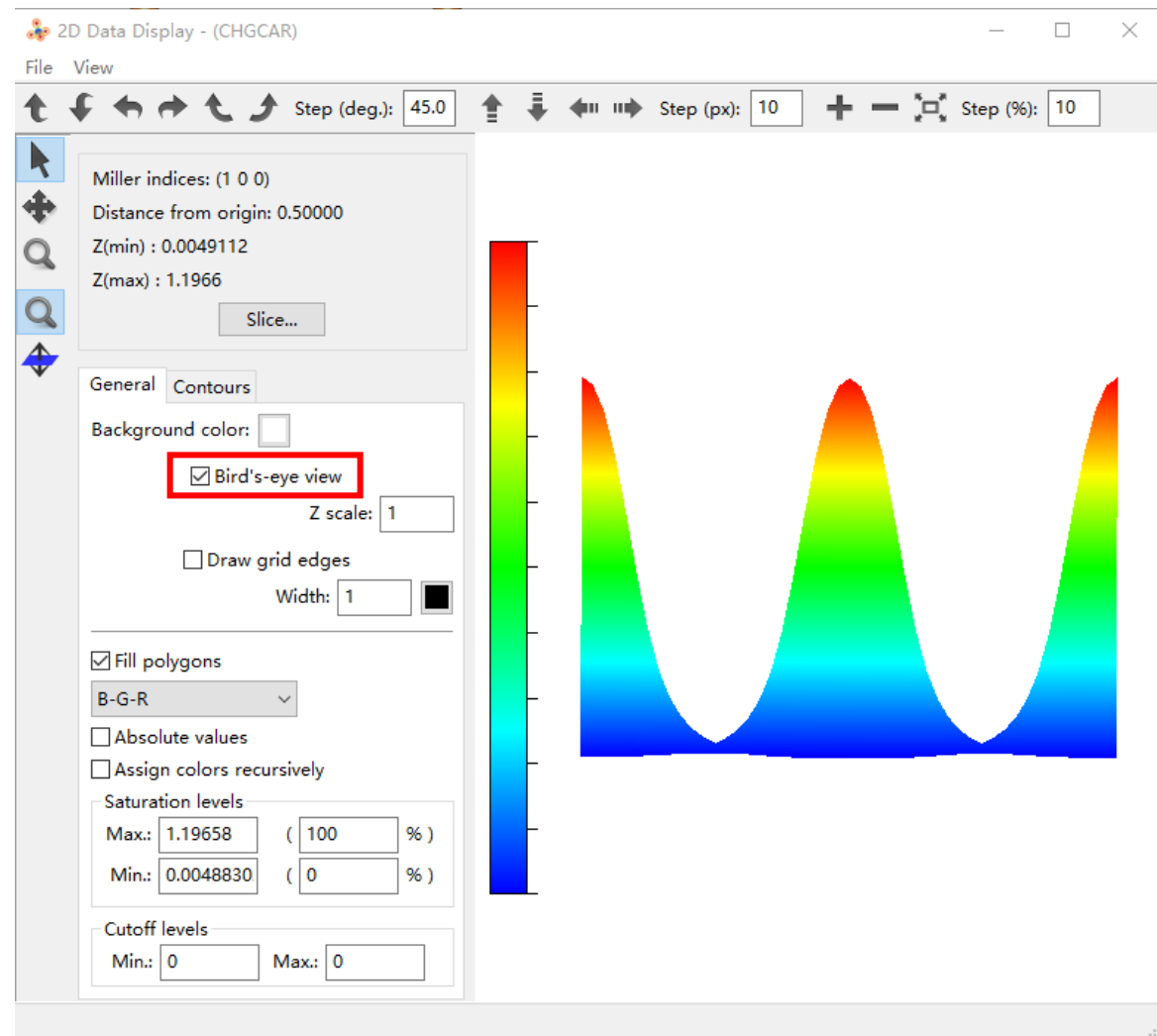
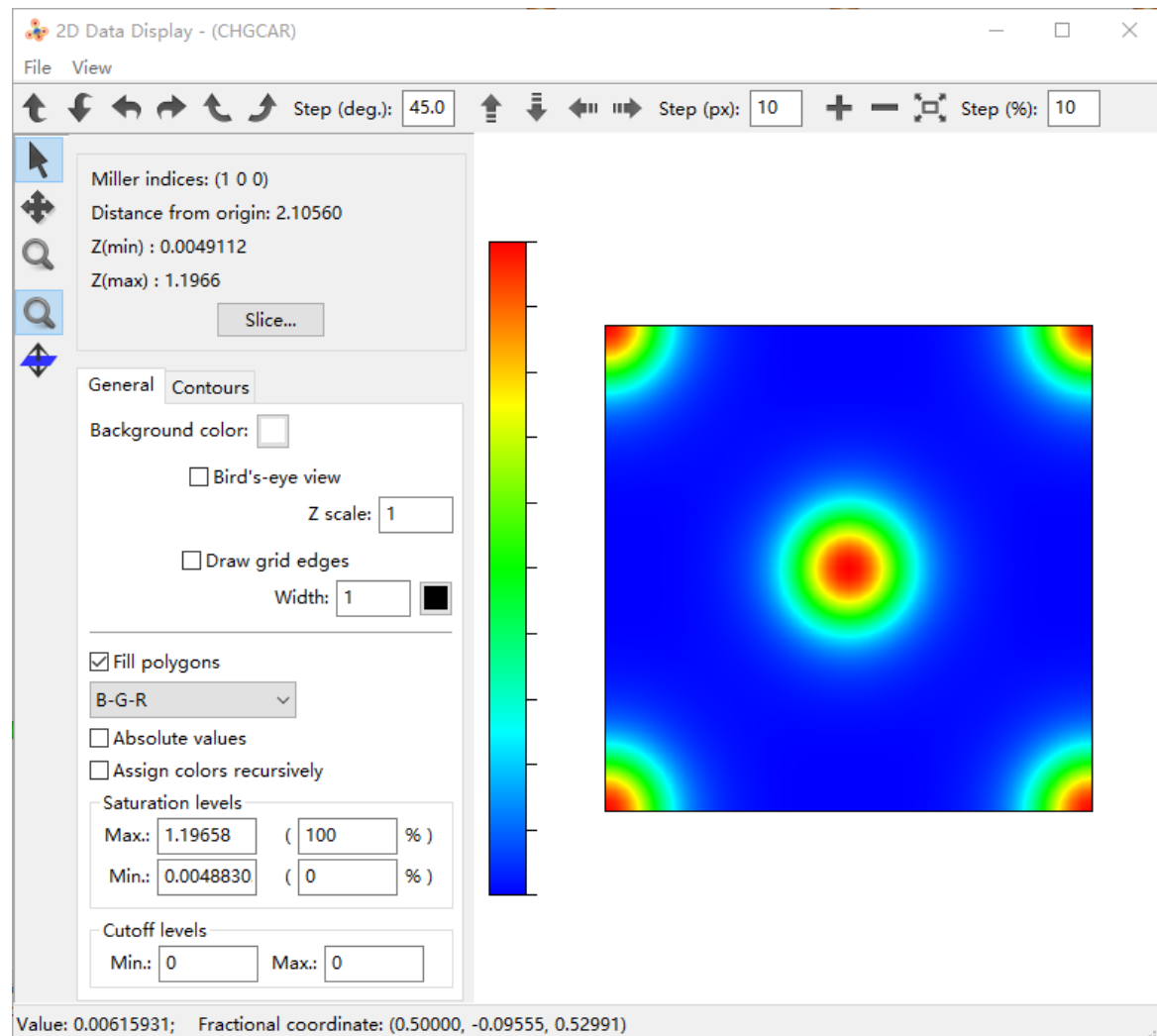
Step 1: Utilities Menu
The "Utilities" menu is open, showing options like "Equivalent Positions...", "Geometrical Parameters...", "Standardization of Crystal Data", "Niggli-Reduced Cell", "Site Potentials and Madelung Energy...", "Powder Diffraction Pattern...", "Structure Factors...", "Fourier Synthesis...", "Model Electron Densities...", "Model Nuclear Densities...", "Patterson Densities", "2D Data Display...", "Line Profile...", "Peak Search...", "Voronoi Integration of Densities...", and "Conversion of Electron Densities...". The "2D Data Display..." option is highlighted.

Step 2: 2D Data Display - (CHGCAR) Window
The "2D Data Display - (CHGCAR)" window is shown. It displays a color map of the electron density. A color bar on the right indicates the mapping from color to numerical values. The "Miller indices: (1 0 0)" are displayed. The "Distance from origin: 2.10560" is shown. The "Z(min): 0.0049112" and "Z(max): 1.1966" are displayed. The "Slice..." button is highlighted with a red box.

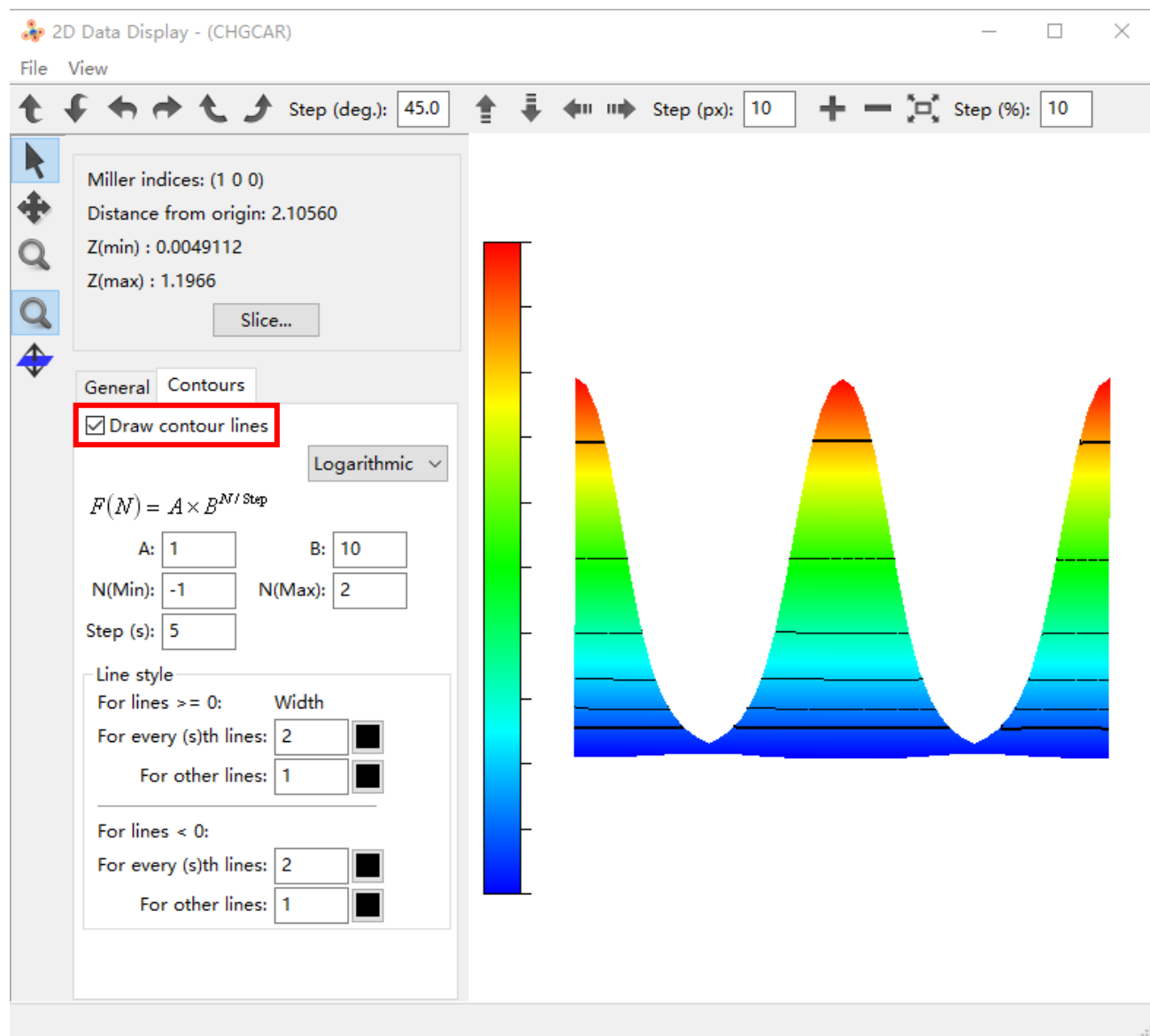
Step 3: Slice Properties Dialog
The "Slice Properties" dialog is open. It shows the "Phase" set to "u". The "Style" is set to "(hkl) plane in the bounding box". The "Miller indices" are set to "h 1", "k 0", and "l 0". The "Distance from origin" is set to "2.1056" Å. The "Ranges of fractional coordinates" are set to "x(min) = 0", "x(max) = 1", "y(min) = 0", "y(max) = 1", "z(min) = 0", and "z(max) = 1". The "Calculate the best plane for the selected atoms" button is highlighted with a red box. The "Set current orientation matrix in the 3D Graphics Area" button is also visible. The "OK", "Cancel", and "Apply" buttons are at the bottom.

Annotations:
- "米勒指数" (Miller indices) is written in red text next to the h, k, l input fields.
- "选择三个原子" (Select three atoms) is written in red text below the "Calculate the best plane for the selected atoms" button.

绘制三维图



加入等高线

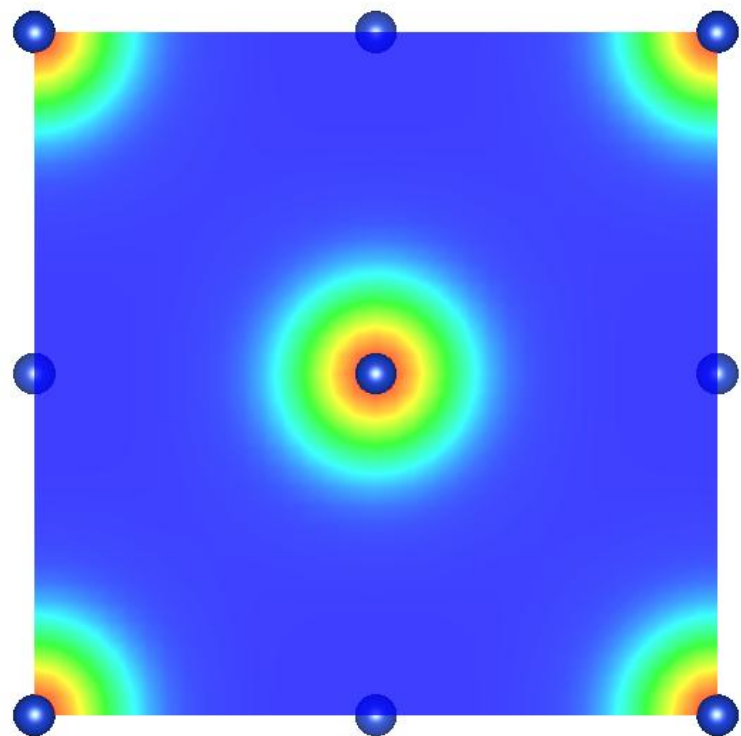


总结

方法一：

优势：存在原子，

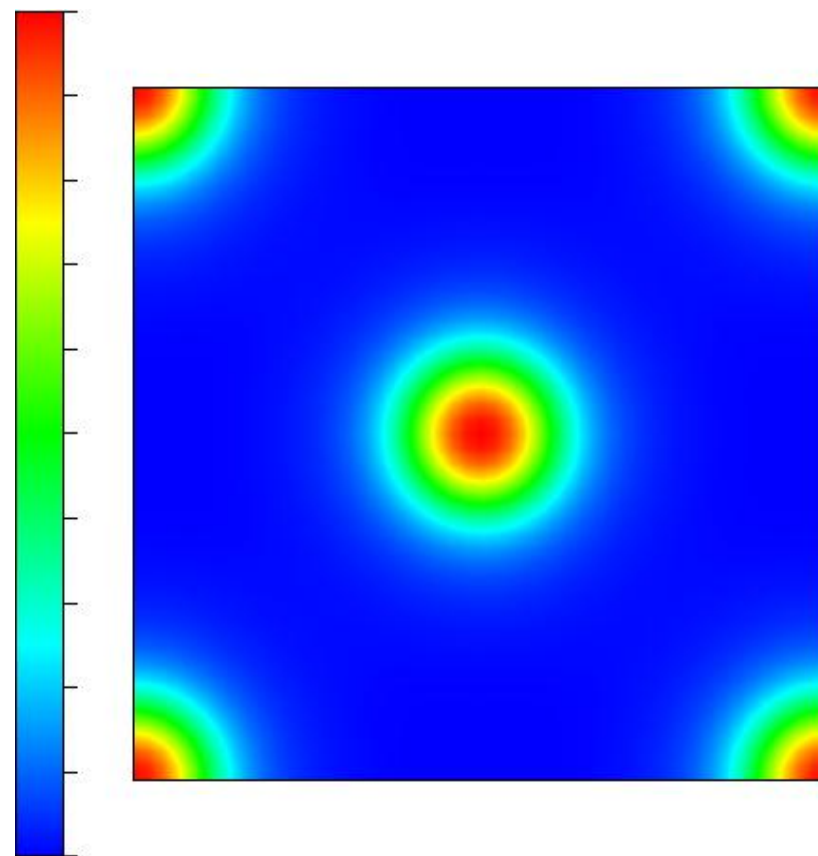
不足：没有colorbar



方法二：

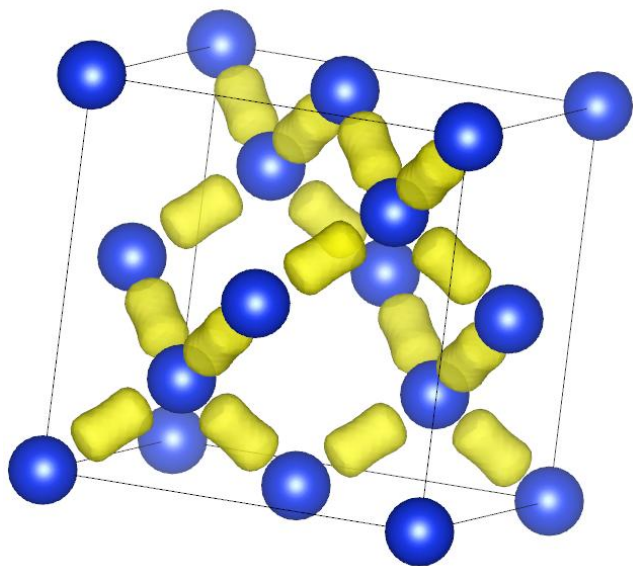
优势：有colorbar

不足：没有原子

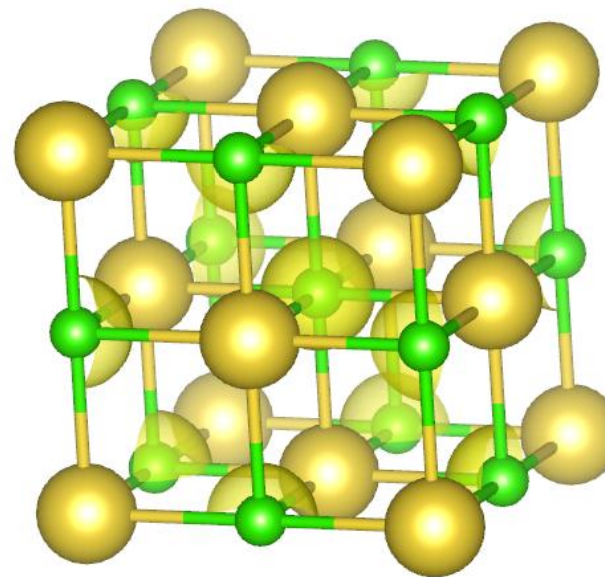


用方法一或方法二，个人取舍

应用举例

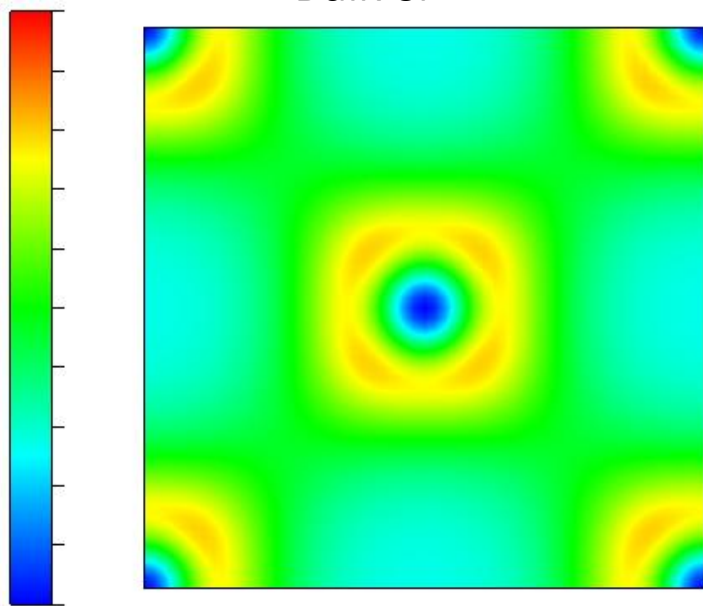


Bulk Si

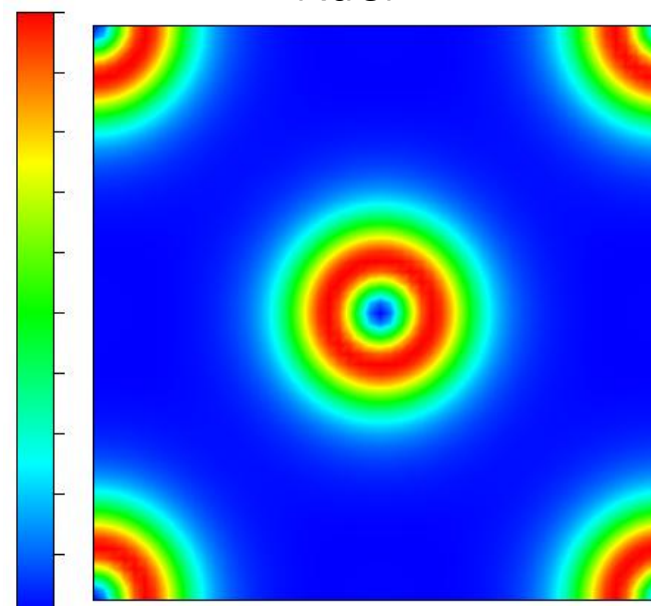


NaCl

Si-Si之间有电荷分布，共用电子对→共价键

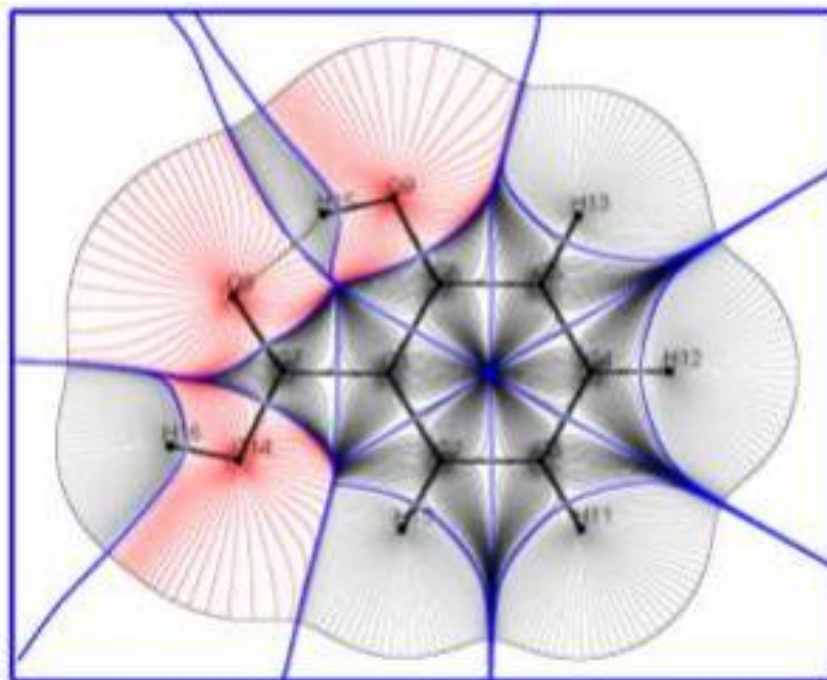


离子键没有明显的成键取向



Bader电荷分析

- Bader电荷亦称为AIM (atomic-in-molecule) 电荷。将电子密度的零通量面作为分界面，划分原子空间的方法，用一套人为的规则定义每个原子上所带的电荷，原子电荷常常用于定性分析



Henkelman课题组开发的Bader程序可以处理VASP的CHGCAR得到bader电荷。

下载地址: <http://theory.cm.utexas.edu/henkelman/code/bader>

Bader电荷分析

- 在优化好的结构基础上，VASP添加关键字：LAECHG=.TRUE. LCHARG=T, NSW=0/1

- AECCAR0：芯电子密度
- AECCAR1：原始原子价电子密度
- AECCAR2：SCF收敛以后的价电子密度

开始会产生 AECCAR0 AECCAR1 两个文件，分别是核电子密度(赝势)，原始原子价电子密度。计算结束会再生成一个AECCAR2文件，是SCF收敛以后的价电子密度。要得到体系的总电荷密度需要将AECCAR0 AECCAR2 两个文件的密度叠加

- **Bader后处理** chgsum.pl路径：/public3/home/a2s000087/bin/vtstscripts-1033

chgsum.pl AECCAR0 AECCAR2, 生成 CHGCAR_sum

bader CHGCAR -ref CHGCAR_sum, 计算bader

- **运行成功生成 ACF.dat BCF.dat AVF.dat**

Bader电荷分析

ACF

笛卡尔坐标

价电子数

原子体积

	X	Y	Z	CHARGE	MIN DIST	ATOMIC VOL
1	1.4016	0.8092	0.0000	10.0309	1.2788	35.4138
2	-0.0485	1.6272	2.3255	10.0361	1.2505	16.1410
3	2.7161	0.0175	4.7261	9.8363	1.0227	14.7246
4	1.2923	0.8347	6.1271	3.2519	0.3168	5.8619
5	1.2962	0.8346	7.3075	6.8448	0.7281	27.0617
VACUUM CHARGE:			0.0000			
VACUUM VOLUME:			0.0000			
NUMBER OF ELECTRONS:			40.0000			

_h, _s	hard, soft
_pv, _sv, _d	包含更内层的电子
_2, _3	含不同价电子数（离子）
_GW	GW计算用
H1.25, H1.5, H.5	赝氢

原子核到一定密度表面的最小距离，可以用来近似衡量离子半径

head POSCAR查看POSCAR中元素排布顺序

grep ZVAL POTCAR 查看对应元素的价电子数目

得失电子数=POTCAR里的相应原子的价电子数目-ACF里的CHGCAR这一列的数值

若为正，则失去电子

若为负，则得到电子

差分电荷charge density difference

差分电荷charge density difference

目的：分析吸附前后的电子密度差别，局部增加或减少

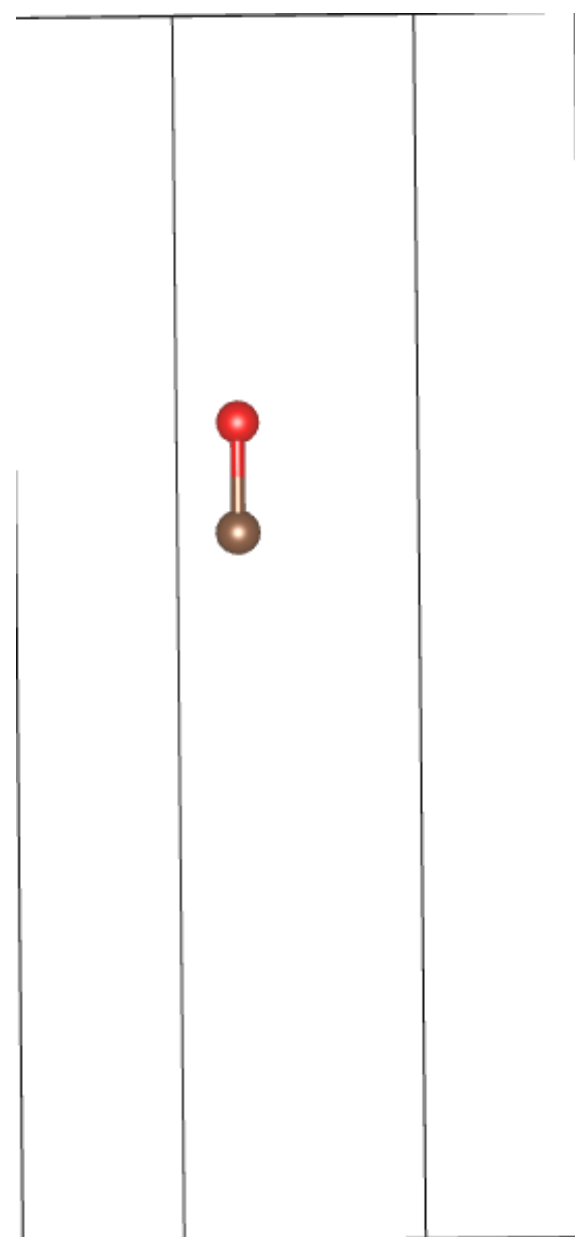
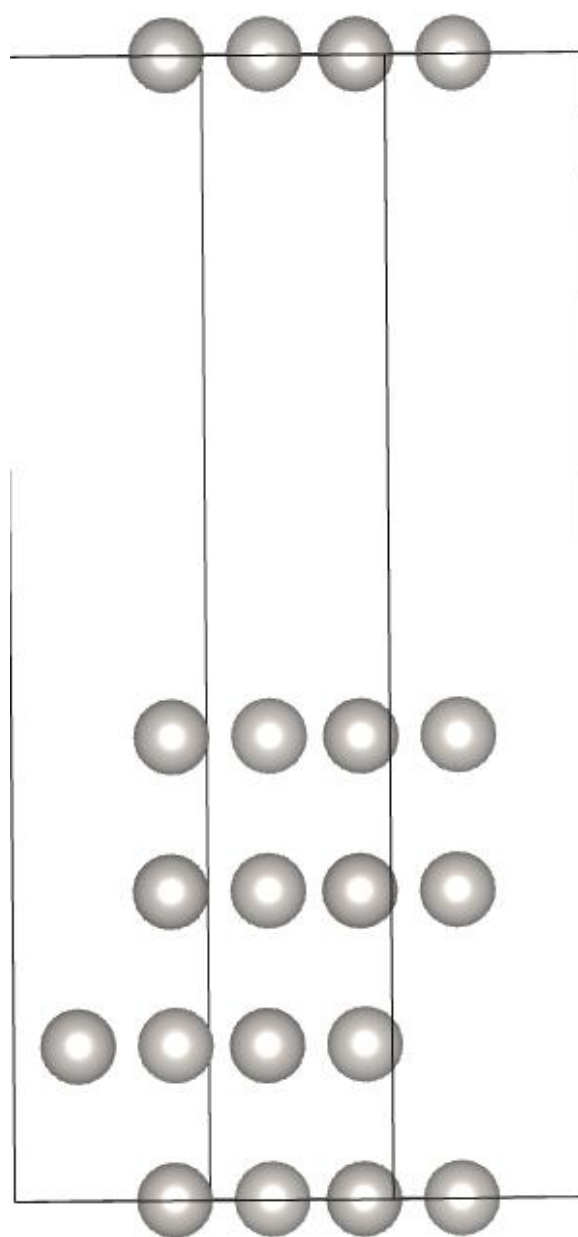
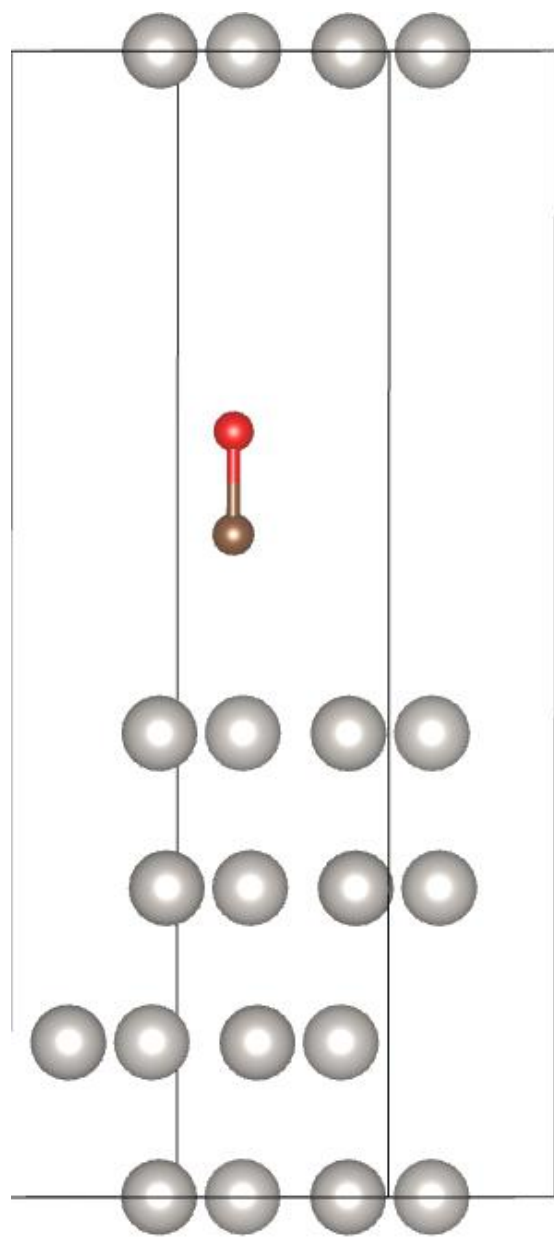
- 体系的电荷密度减去组成它的两个或几个片段的密度
- 只需静态计算，并打开CHGCAR
- NSW=0/1, LCHARG=T

$$\Delta\rho = \rho_{AB} - \rho_A - \rho_B$$

$$\Delta\rho = \rho_{ABC} - \rho_A - \rho_B - \rho_C$$

以CO吸附在Pt (111) 上为例

- 步骤一：优化 $\text{CO/Pt (111)}_{\text{top}}$ 的结构，计算电荷密度；
 - 步骤二：分别计算 Pt (111) 和 CO 的电荷密度
 - 步骤三：用 `vaspkit 314` 功能做电荷差分。依次输入三个片段的 CHGCAR 路径
-
- 注意：只步骤一需优化，后面的计算关闭LSIF, 所有的计算参数保持不变



分别计算三种结构的电荷密度，得到相应CHGCAR

1. 三个文件夹都包含各自CHGCAR

```
[temp@mgt ptco]$ ls  
co  pt  ptco  
[temp@mgt ptco]$
```

2. vaspkit

```
[temp@mgt ptco]$ vaspkit  
  
      \\\//  
      /  (o)(o)  \  
      (| (o)(o) |)  
Hey, you must know what you are doing.  
Otherwise you might get wrong results.  
+-----o00o--()--o00o-----+  
|                               |  
|      VASPKIT Version 1.2.5 (08 Feb. 2021)      |  
|      Core Developer: Vei WANG (wangvei@icloud.com)      |  
|      Main Contributors: Nan XU, Jin-Cheng LIU & Gang TANG      |  
|      Online Tutorials Available on Website: https://vaspkit.com      |  
+-----o00o-----+  
|      ( )      Oooo.      VASPKIT Made Simple      |  
|      \ (      ( )      |  
|      \_)      ) /      |  
|      (/_      (/_      |  
===== Structural Options =====  
1) VASP Input Files          2) Elastic Properties  
3) K-Path for Band-Structure  4) Structure Editor  
5) Catalysis-ElectroChem Kit  6) Symmetry Search  
8) Advanced Structure Models  
===== Electronic Options =====  
11) Density-of-States        21) DFT Band-Structure  
23) 3D Band-Structure        25) Hybrid-DFT Band-Structure  
26) Fermi-Surface            28) Band-Structure Unfolding  
31) Charge & Spin Density     42) Potential Analysis  
51) Wave-Function Analysis    62) Magnetic Properties  
===== Various Utilities =====  
71) Optical-Properties        72) Molecular-Dynamics Kit  
73) VASP2other Interface      74) User Interface  
91) Semiconductor Kit         92) 2D-Material Kit  
0) Quit  
----->>
```

3. 输入314

```
----->>
314
===== File Options =====
Input the Names of Charge/Potential Files with Space:
(e.g., to get AB-A-B, type: ~/AB/CHGCAR ./A/CHGCAR ../B/CHGCAR)
```

4. 输入CHGCAR所在路径

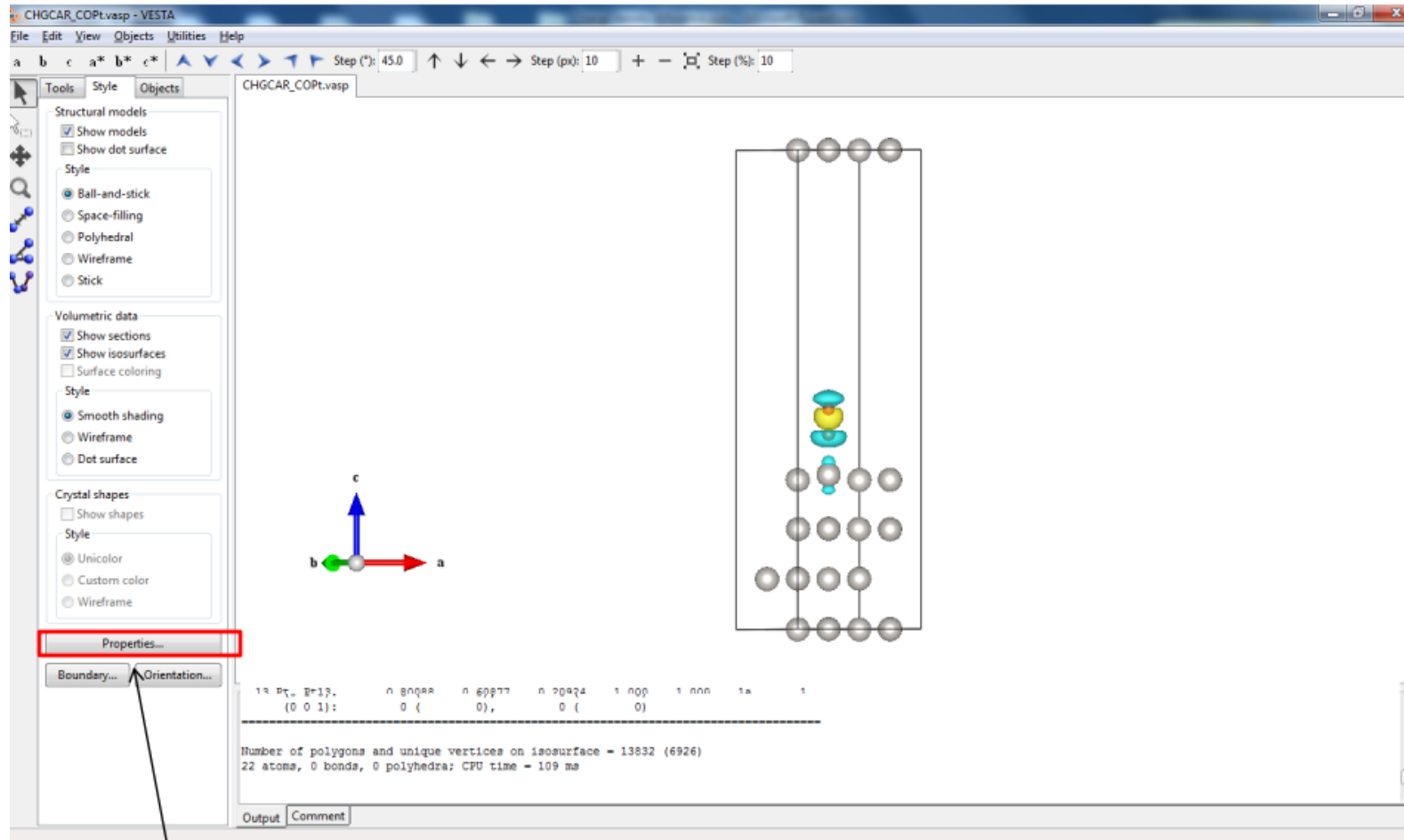
```
----->>
./ptco/CHGCAR ./pt/CHGCAR ./co/CHGCAR

-->> (01) Reading Structural Parameters from ./ptco/CHGCAR File..
-->> (02) Reading Charge Density From ./ptco/CHGCAR File...
-->> (03) Reading Structural Parameters from ./pt/CHGCAR File...
-->> (04) Reading Charge Density From ./pt/CHGCAR File...
-->> (05) Reading Structural Parameters from ./co/CHGCAR File...
-->> (06) Reading Charge Density From ./co/CHGCAR File...
-->> (07) Written CHGDIFF.vasp File!

+-----+
|               * ACKNOWLEDGMENTS *               |
| Other Contributors (in no particular order): Peng-Fei LIU, |
| Xue-Fei LIU, Dao-Xiong WU, Zhao-Fu ZHANG, Tian WANG, Qiang LI, |
| Ya-Chao LIU, Jiang-Shan ZHAO, Qi-Jing ZHENG, iGo Li and You! |
| Advisors: Wen-Tong GENG, Yoshiyuki KAWAZOE |
+-----+
|               * CITATIONS *               |
| If you have used VASPKIT in your work, PLEASE cite: |
| [1] V. WANG, N. XU, J.-C. LIU, G. TANG, W.-T. GENG, VASPKIT: A |
| User-Friendly Interface Facilitating High-Throughput Computing |
| and Analysis Using VASP Code, arXiv:1908.08269 (2019) |
+-----+
[temp@ngt ptco]$ ls
CHGDIFF.vasp  co  pt  ptco
```

得到的CHGDIFF.vasp用VESTA打开，等值面 (isosurface) 蓝色部分电荷密度减小，黄色密度电荷密度增加

The isosurface of charge density difference is obtained.



Click Properties... to modify the isosurface.

修改isosurface

调整等值面

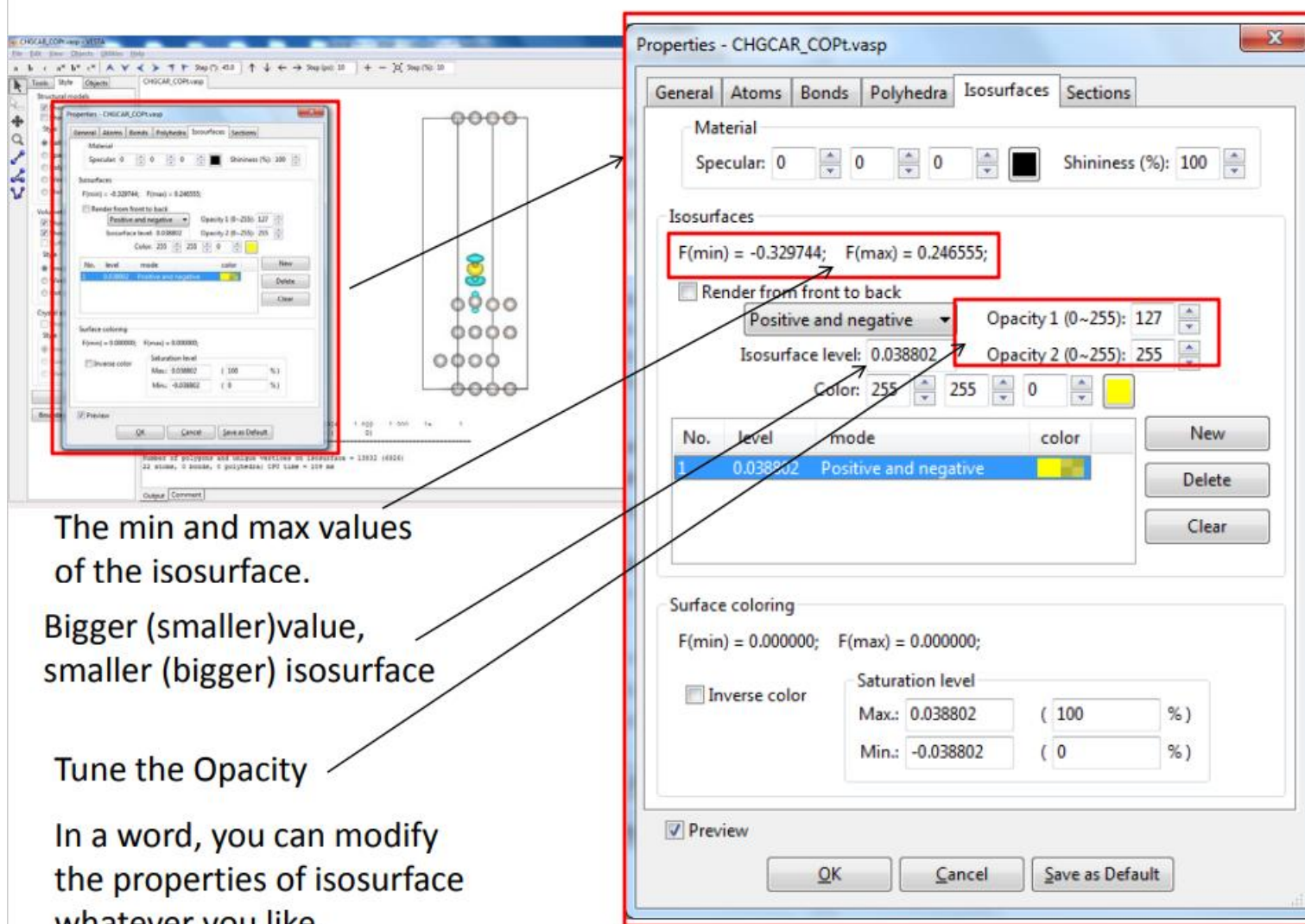
The min and max values of the isosurface.

Bigger (smaller)value, smaller (bigger) isosurface

修改不透明度

Tune the Opacity

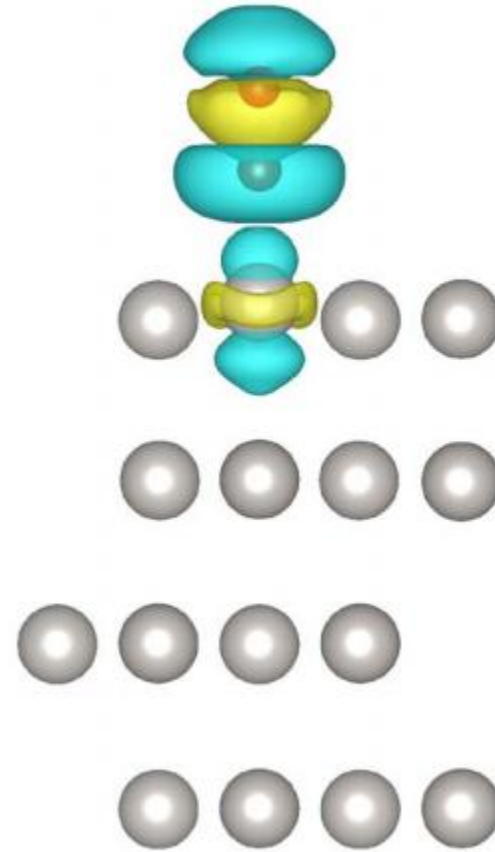
In a word, you can modify the properties of isosurface whatever you like.



目前为止，得到的图是我们常见的图，没有colorbar

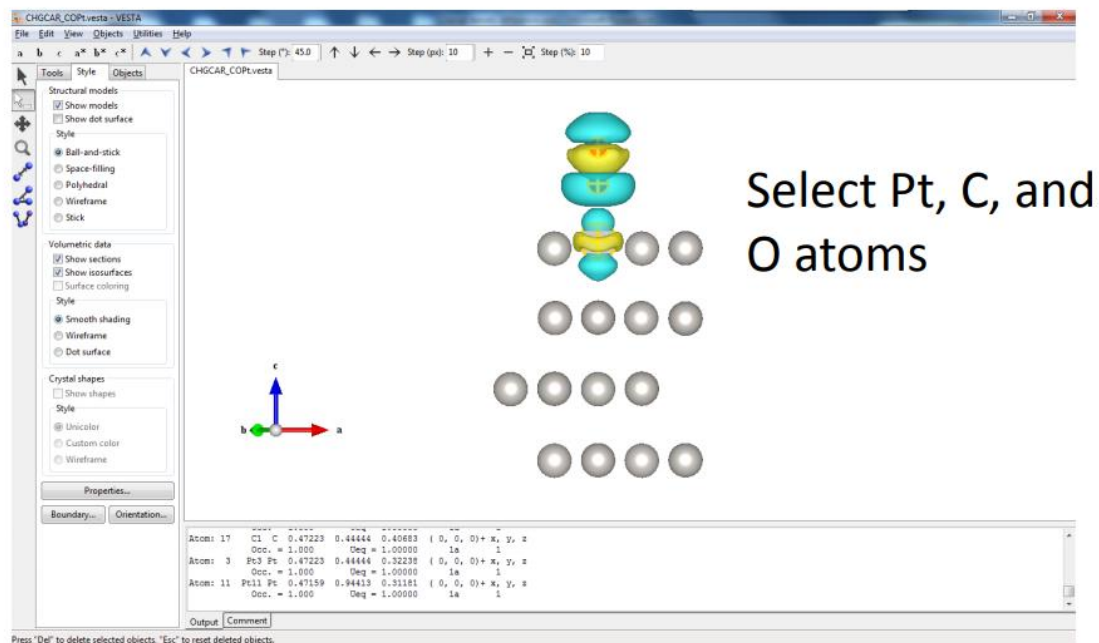
The final figure

This is the side view of the isosurface. You can also export other views of the isosurface.



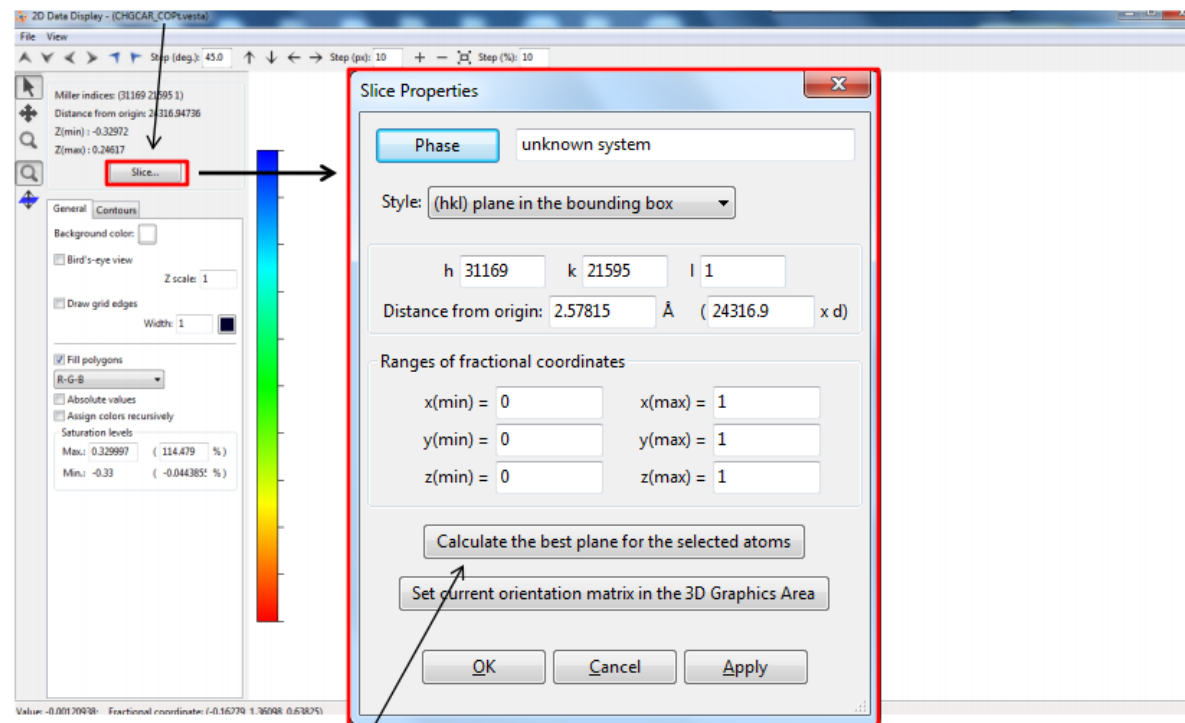
如果想画有colorbar的，首先选中想要看电荷密度变化的那个面或者线上存在的三个原子
(见前面的三个原子确定一个平面)

2D slice



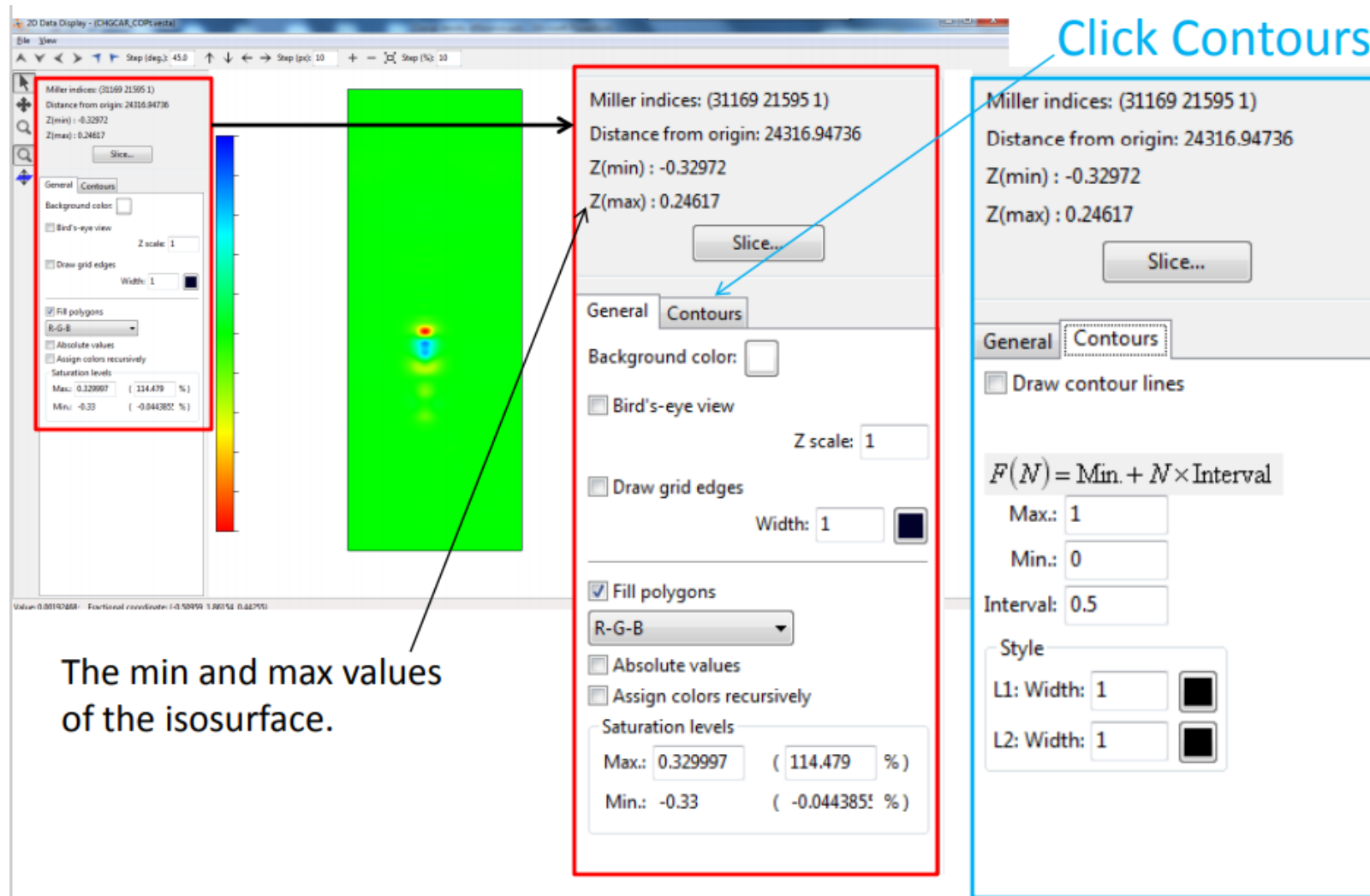
Utilities → 2D Data Display

Click Slice



Click Calculate the best plane for the selected atoms, and then OK.

等高线

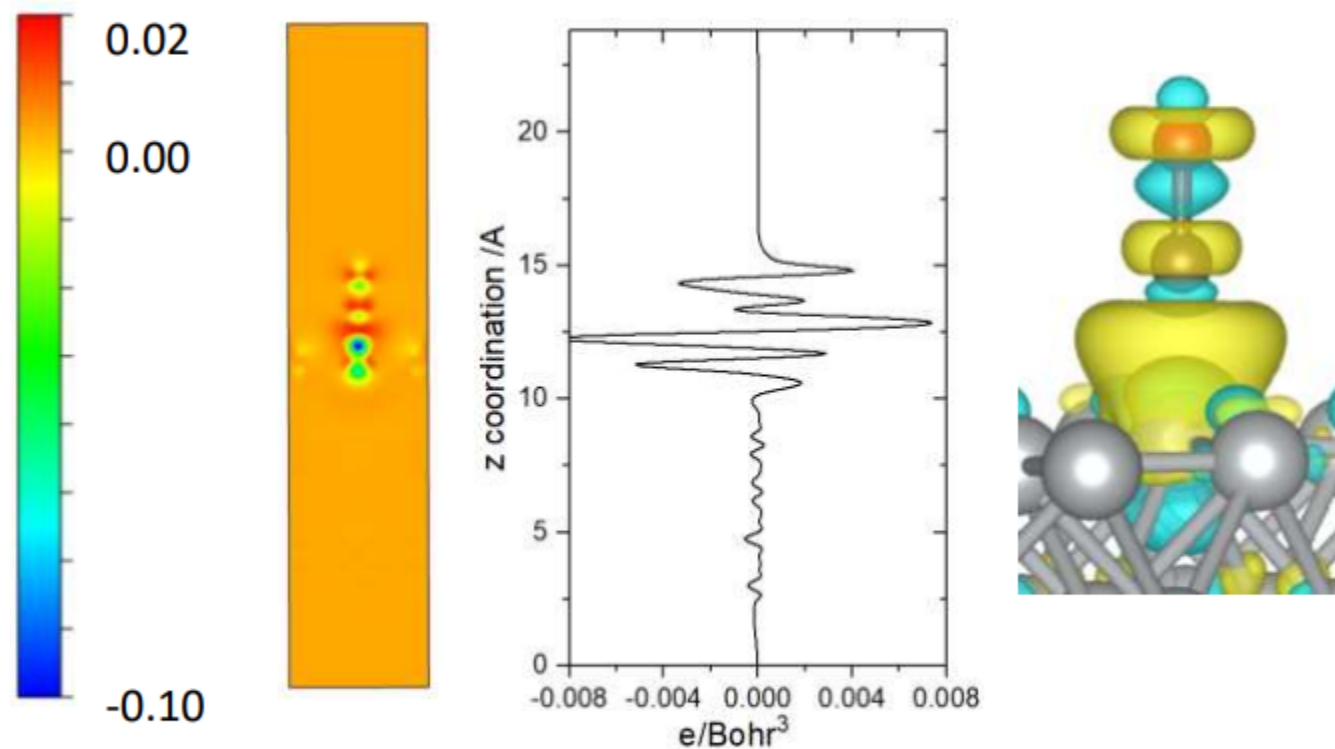


The min and max values of the isosurface.

Tune the display of the figure whatever you like by using these two boxes.

得到的CHGDIFF.vasp用vesta打开， isosurface青色部分电荷密度减小，黄色密度电荷密度增加。

CO/Ni(100)电荷差分， $\rho = \rho(\text{CO/Ni}) - \rho(\text{CO}) - \rho(\text{Ni})$

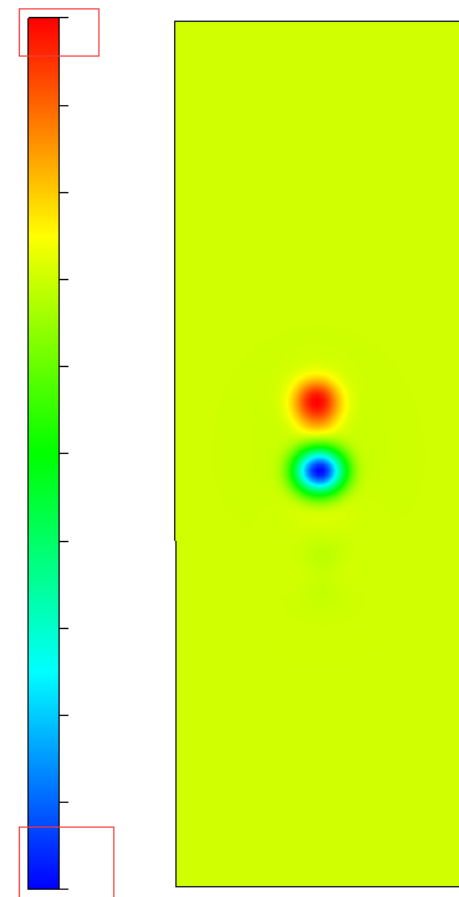
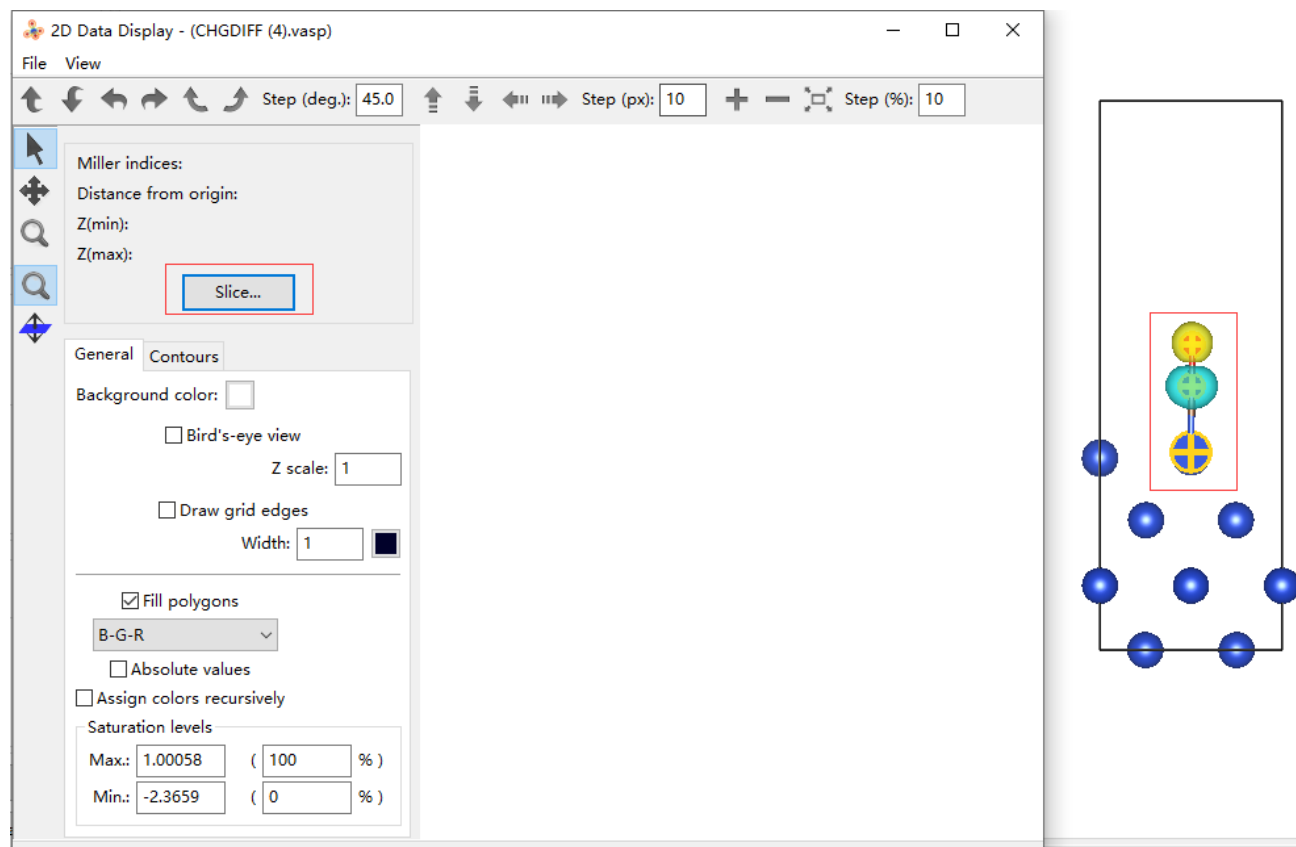
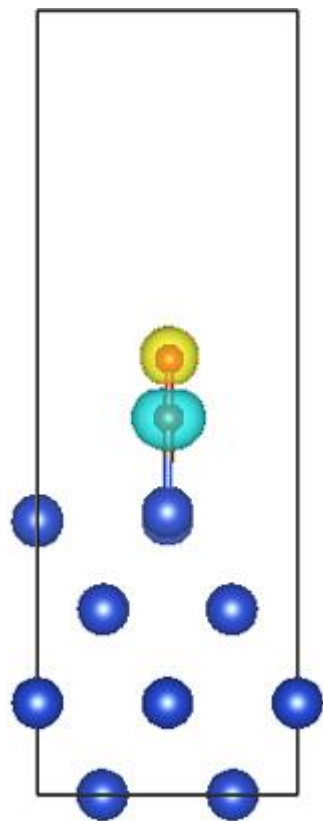


一维差分密度图 步骤四：把上一步CHGDIFF.vasp在新的文件夹中重命名为CHGCAR，运行vaspkit功能316。得到CHGPAVG.dat，直接导入到origin里作图，把X-Y坐标反转一下即可。

例子：Pt(111)吸附CO

先选中想要看电子
态密度变化的那个
面上的三个原子

前后阈值去
properties里的
section模块查看



步骤四：把上一步CHGDIFF.vasp在新的文件夹中重命名为CHGCAR，运行vaspkit功能316。得到CHGPAVG.dat，直接导入到origin里作图，把X-Y坐标反转一下即可。

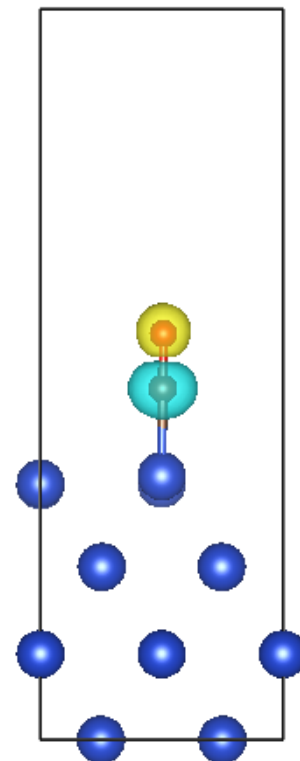
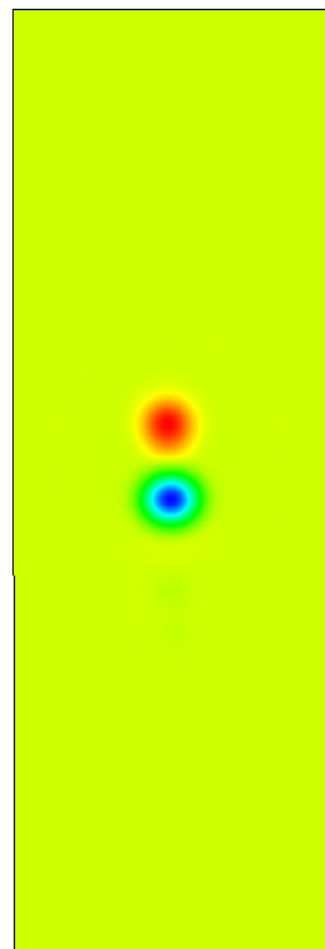
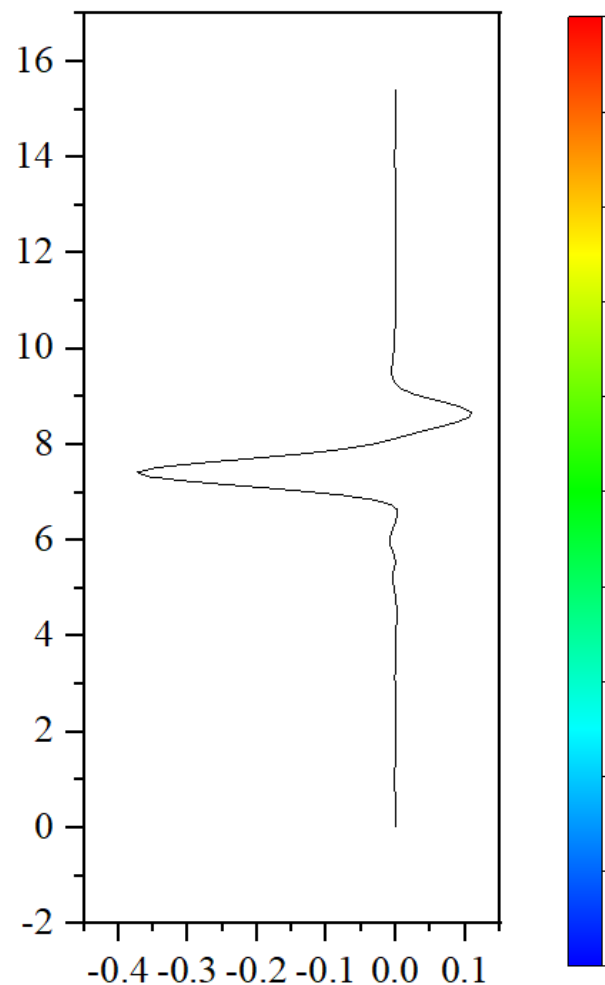
```
[temp@mgt test]$ ls
CHGCAR
[temp@mgt test]$ vaspkit
```

```
316
===== Charge Density File =====
1) CHGCAR
2) CHGDIFF.vasp
3) Specified Charge Density File

0) Quit
9) Back
----->>
1
+----- Warm Tips -----+
+----- You MUST Know What You are doing -----+
+----- Check Convergence of Planar Average Value on Vacuum-Thickness! -----+
+----- Specified Direction -----+
===== Specified Direction =====
Which Direction is Selected to Calculate Planar Averaging?
1) Lattice a Direction
2) Lattice b Direction
3) Lattice c Direction

0) Quit
9) Back
----->>
3
-->> (01) Reading Structural Parameters from CHGCAR File...
-->> (02) Reading Charge Density From CHGCAR File...
-->> (03) Written PLANAR_AVERAGE.dat File!
```

```
[temp@mgt test]$ ls
CHGCAR PLANAR_AVERAGE.dat
```

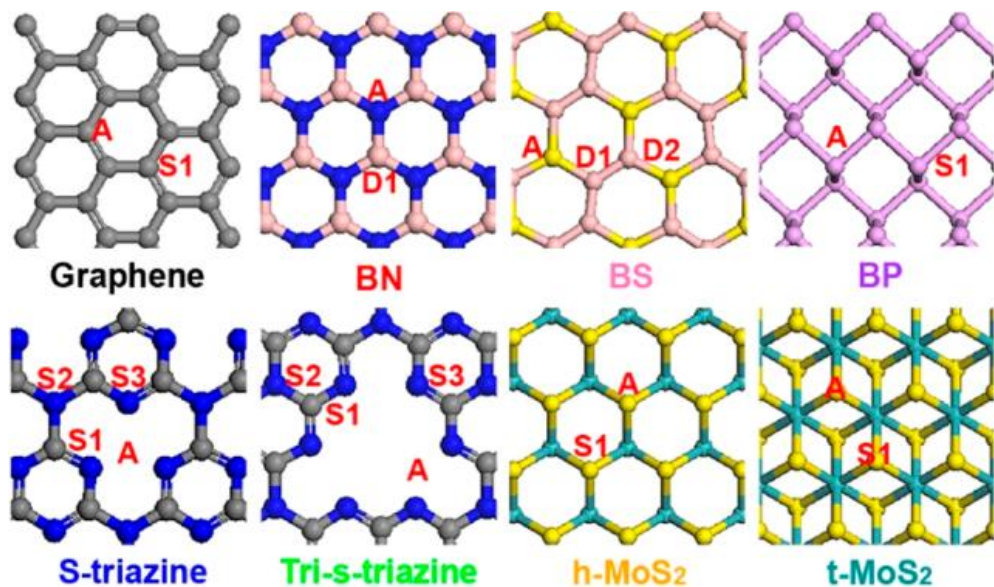


吉布斯自由能计算 (频率计算)

吉布斯自由能应用案例：催化性能评估

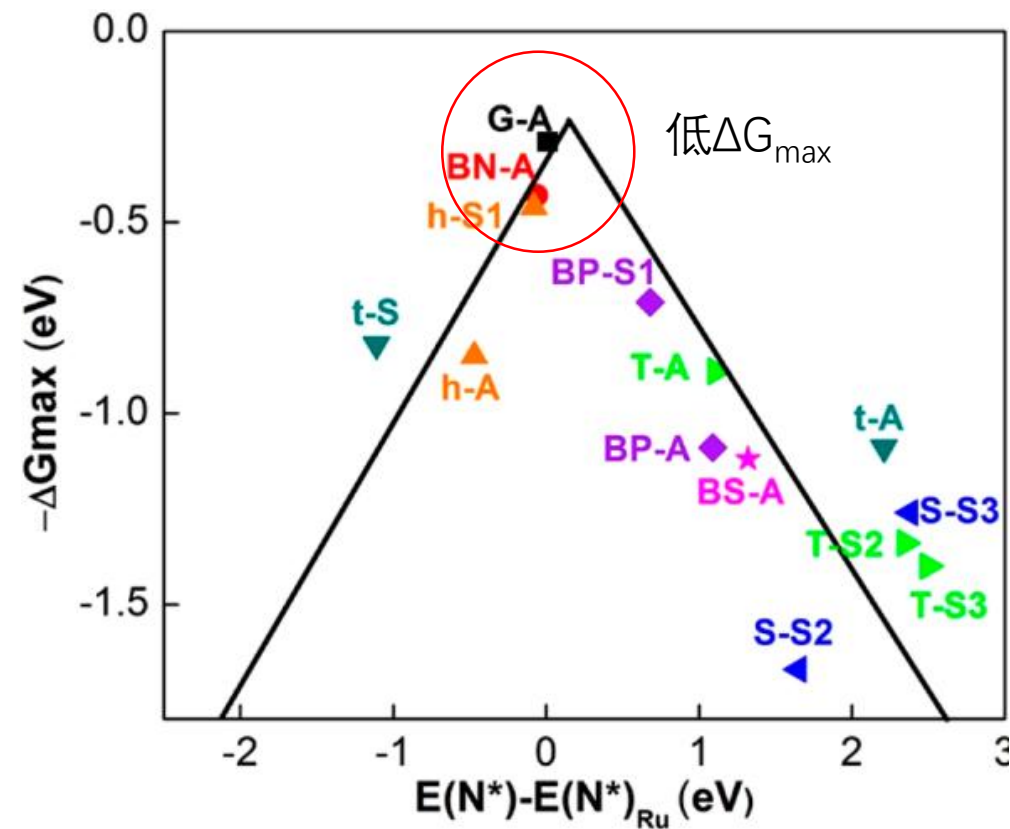
单原子硼作为p-区元素催化剂在氮还原反应（NRR）中的应用

氮还原反应(NRR)火山图



A: absorbed
S: substituted
D: lattice

B吸附在二维基底材料表面
B替换原本的原子
晶格B



$E(N^*)$: binding energy of a single N
 $E(N^*)_{\text{Ru}}$: N-adsorption on Ru as a reference
 $\Delta G_{\max}$: largest value in elementary steps

G-A的氮还原反应(NRR)反应路径和台阶图

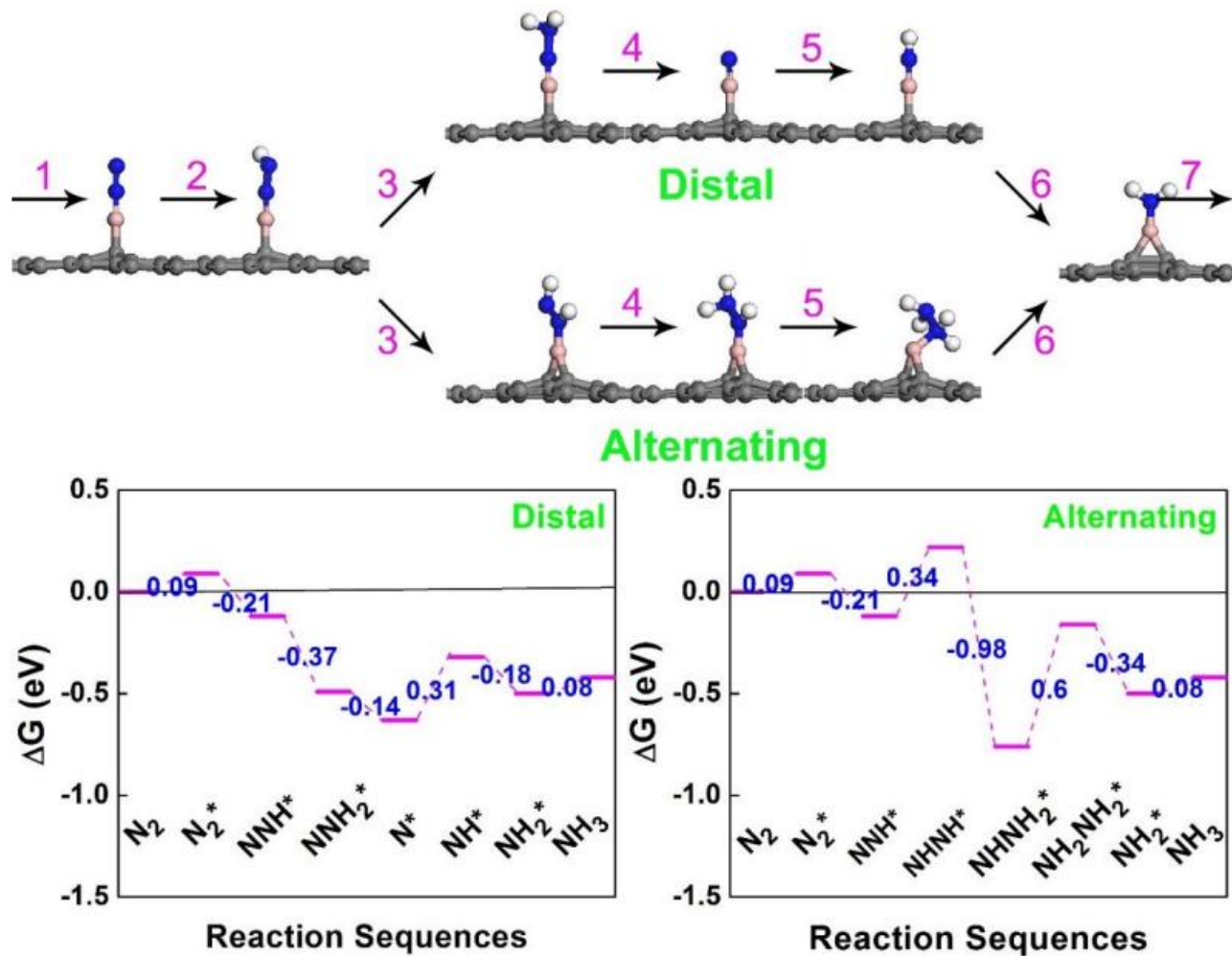
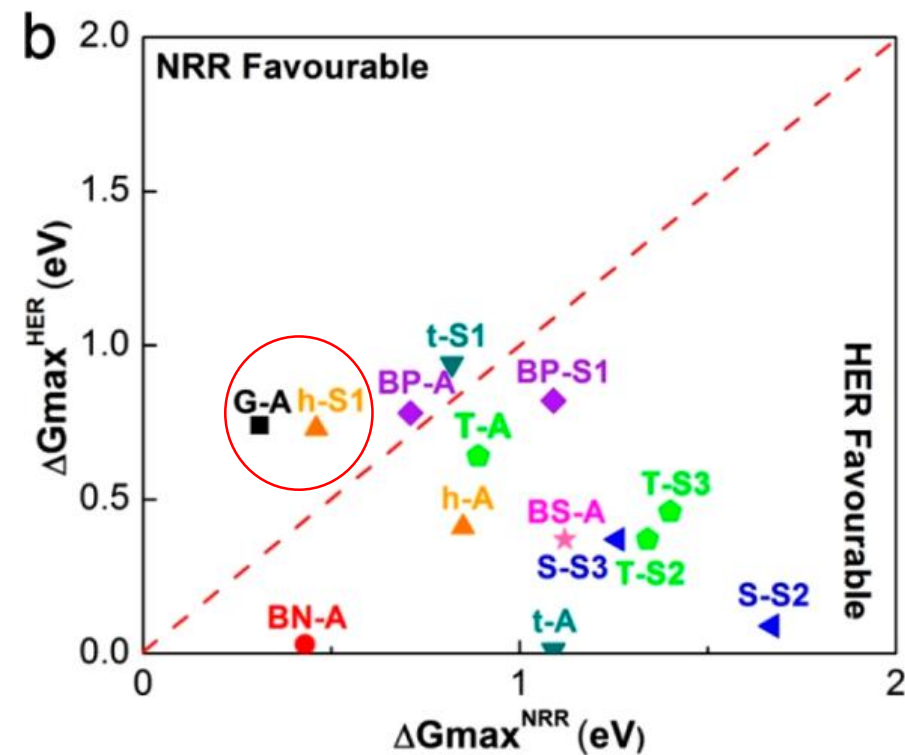


Figure S9. Optimized geometric structures of various intermediates along the distal and alternating paths of NRR on boron atom adsorbed on perfect graphene. And calculated reaction ΔG of two reaction pathways.

Distal $\Delta G_{max}=0.31$ eV

副反应：析氢反应(HER)

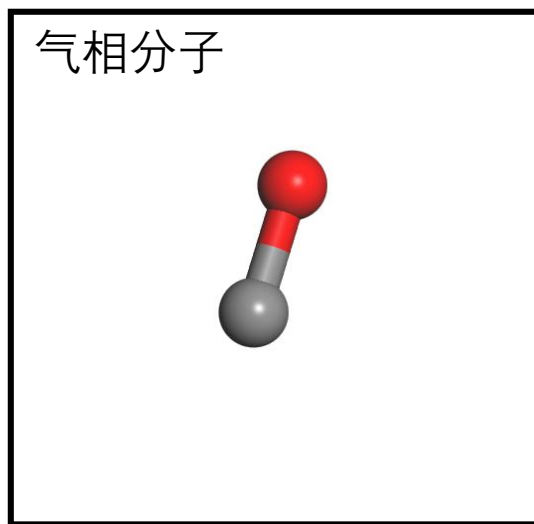


催化剂两大性能指标:

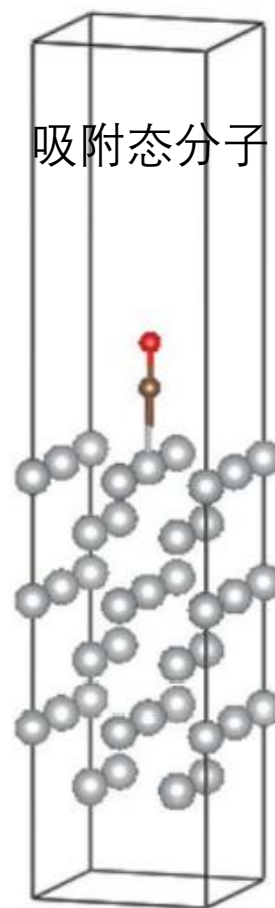
1. ΔG_{max} (反应能垒), 需要电压驱动
2. FE: 实际产率与理论产率比(需遏制副反应)

自由能校正原理

无论是气相化学反应，还是表面、材料的化学变化，都是在一定反应条件下进行的，但是直接用第一性原理程序计算出来的都是指在0K下忽略**零点振动能(ZPE)**的能量(俗称电子能量)。自由能校正中首先要计算配分函数，然后校正体系内能(U)，焓(H)，熵(S)，ZPE。由此可以引入，反应温度、压力对能量的影响，从而可以进一步考虑 pH、电极电势等等外界因素对化学势的影响，讨论结果才和真实的实验情况更为接近



3个自由度
振动 (vibration)
平动 (translation)
转动 (rotation)



3个自由度
振动 (vibration)
平动 (frustrated translation)
转动 (frustrated rotation)

一般会将吸附分子的平动和转动这一部分贡献归到振动里，也就是说表面吸附分子的3N个振动(虚频除外)都用于计算热力学量的校正

$$G = E_{\text{DFT}} + E_{\text{ZPE}} + \int C_p dT - TS$$

$$E_{\text{ZPE}} = \frac{1}{2} \sum_i h \nu_i$$

$$Z = \sum_{n=0}^{\infty} e^{-\beta E_n} \quad S = k_B \left[\frac{\partial}{\partial T} (T \ln Z) \right]_V$$

不同的振动频率对熵的贡献

(d) The vibrational contribution

The vibrational contribution to the molar entropy, S_m^V , is obtained by combining the expression for the molecular partition function (eqn 15B.15, $q^V = 1/(1 - e^{-\beta\epsilon})$) with the expression for the mean energy (eqn 15C.8, $\langle \epsilon^V \rangle = \epsilon / (e^{\beta\epsilon} - 1)$), to obtain

$$S_m^V = \frac{\overbrace{U_m - U_m(0)}^{N_A \langle \epsilon^V \rangle}}{\underbrace{T}_{1/k\beta}} + R \ln q^V = \frac{\overbrace{N_A k \beta \epsilon}^R}{e^{\beta\epsilon} - 1} + R \ln \frac{1}{1 - e^{-\beta\epsilon}}$$

$$= R \left\{ \frac{\beta\epsilon}{e^{\beta\epsilon} - 1} - \ln(1 - e^{-\beta\epsilon}) \right\}$$

Now we recognize that $\epsilon = hc\tilde{\nu}$ and obtain

$$S_m^V = R \left\{ \frac{\beta hc \tilde{\nu}}{e^{\beta hc \tilde{\nu}} - 1} - \ln(1 - e^{-\beta hc \tilde{\nu}}) \right\} \quad \text{Vibrational contribution to the entropy} \quad (15E.13a)$$

Once again it is convenient to express this formula in terms of a characteristic temperature, in this case the vibrational temperature $\theta^V = hc\tilde{\nu}/k$:

$$S_m^V = R \left\{ \frac{\theta^V / T}{e^{\theta^V / T} - 1} - \ln(1 - e^{-\theta^V / T}) \right\} \quad \text{Vibrational contribution to the entropy} \quad (15E.13b)$$

```
qli@p015:~$ for i in
10 33.5696 0.1023
30 24.4416 0.0745
50 20.2071 0.0616
70 17.4286 0.0531
90 15.3643 0.0468
110 13.7274 0.0418
130 12.3760 0.0377
150 11.2299 0.0342
170 10.2387 0.0312
190 9.3692 0.0285
210 8.5980 0.0262
230 7.9080 0.0241
250 7.2864 0.0222
270 6.7233 0.0205
290 6.2109 0.0189
310 5.7429 0.0175
330 5.3141 0.0162
350 4.9202 0.0150
370 4.5576 0.0139
390 4.2233 0.0129
410 3.9145 0.0119
430 3.6290 0.0111
450 3.3648 0.0102
470 3.1201 0.0095
490 2.8932 0.0088
```

波数 熵 TS

频率越小，熵越大，对TS贡献越大，所以前面说的受阻平动转动对熵的影响很大，所以气相分子和吸附态的分子的TS得分开计算

吉布斯自由能计算: INCAR

1. **IBRION = 5** (告诉服务器要计算振动频率)

POTIM = 0.015 (离子移动的步长)

NFREE = 2 (表明原子在某一方向上正反两个方向移动)

NSW = 1 (绝对不能设成0)

一般服务器不支持并行计算频率, 所以不要出现**NPAR NCORE**参数

2. **POSCAR**: 计算吸附分子时, 将表面的原子**固定**住 (放开吸附分子)

3. **KPOINTS POTCAR**不变

4. 提交任务, 频率计算的离子步数是固定的, $N = 6 \times [\text{放开的原子数}] + 1$ 。

比如只放开了一个原子, 那么 $N = 6 \times 1 + 1 = 7$

5. 检查输出文件: 计算频率典型的输出就是每个离子步之后都有振动自由度的信息

grep cm-1 OUTCAR, 一个原子有三个自由度, 即三个振动方式

6. 后处理: **vaspkit 502**(气相分子)/**501**(吸附分子), 提取ZPE和TS

自由能计算案例：气相H₂

第一步：优化结构

第二步：修改INCAR，计算振动频率

第三步：vaspkit 502处理气相分子

```
26) Fermi-Surface          28) Band-Structure Unfolding
31) Charge & Spin Density  42) Potential Analysis
51) Wave-Function Analysis 62) Magnetic Properties
===== Various Utilities =====
71) Optical-Properties      72) Molecular-Dynamics Kit
73) VASP2other Interface   74) User Interface
91) Semiconductor Kit      92) 2D-Material Kit
0) Quit
----->>
502
+----- Warm Tips -----+
  See An Example in vaspkit/examples/thermo_correction/ethanol.
  This Feature Was Contributed by Nan XU, Jincheng Liu and Sobereva.
  Included Vibrations, Translation, Rotation & Electron contributions.
  GAS molecules should not be with any fix.
  -->> (01) Reading Structural Parameters from CONTCAR File...
  -->> (02) Analyzing Molecular Symmetry Information...
  Molecular Symmetry is: D(inf)h
  Linear molecule found!
+-----+

```

第四步：输入温度，压强和自旋多重度得到结果

```
+-----+
Please input Temperature (K)!
298
Please input Pressure (Atm)!
1
Please input Spin multiplicity!--(Number of Unpaired electron + 1)
1
----->>

```

```
Temperature (T)      :      298.0 K
Pressure (P)         :    101325.000 Pa
Spin-Multiplicity (S) :          1
Zero-point energy E_ZPE :      6.190 kcal/mol    0.268417 eV
Thermal correction to U(T):      7.670 kcal/mol    0.332614 eV
Thermal correction to H(T):      8.263 kcal/mol    0.358293 eV
Thermal correction to G(T):     -1.026 kcal/mol   -0.044509 eV
Entropy S            :      130.421 J/(mol*K)    0.001352 eV/K
Entropy contribution T*S :    38865.509 J/(mol)    0.402802 eV

```

Table S3. DFT total energies and zero point energy, enthalpic correction and entropy correction at 298.15 K.

Species	E _{DFT} (eV)	E _{ZPE} (eV)	∫ C _p dT (eV)	-TS (eV)
H ₂ (g)	-6.772	0.276	0.09	-0.403
CO ₂ (g)	-22.958	0.310	0.10	-0.663
CO(g)	-14.779	0.138	0.09	-0.596
H ₂ O(g=l)	-14.222	0.578	0.10	-0.673

频率计算的作用

- 1 确定结构是否稳定(有无虚频);
- 2 看振动方式和大小, 用来和实验对比(红外谱图)
- 3 反应热, 反应能垒, 吸附能等的零点能矫正 (自由能自己算);
- 4 确认过渡态(有一个振动的虚频)
- 5 热力学中计算entropy, 用于计算化学势, 微观动力学中的指前因子和反应能垒。



Home Browse the Journal Articles ASAP Current Issue Submission & Review Open Access

Article

The Entropies of Adsorbed Molecules

Charles T. Campbell* and Jason R. V. Sellers
Department of Chemistry, University of Washington, Seattle, Washington 98195-1700, United States

J. Am. Chem. Soc., **2012**, 134 (43), pp 18109–18115
DOI: 10.1021/ja3080117
Publication Date (Web): October 3, 2012
Copyright © 2012 American Chemical Society
campbell@chem.washington.edu

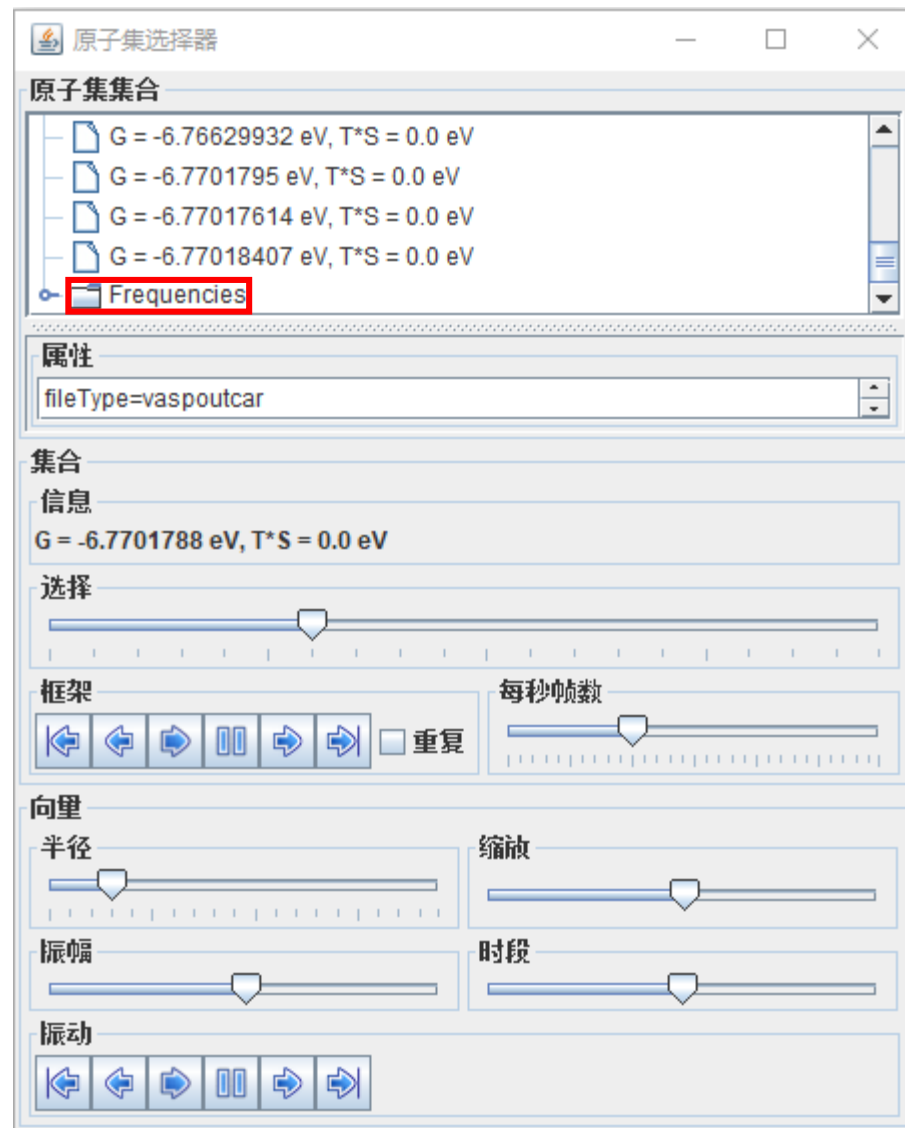
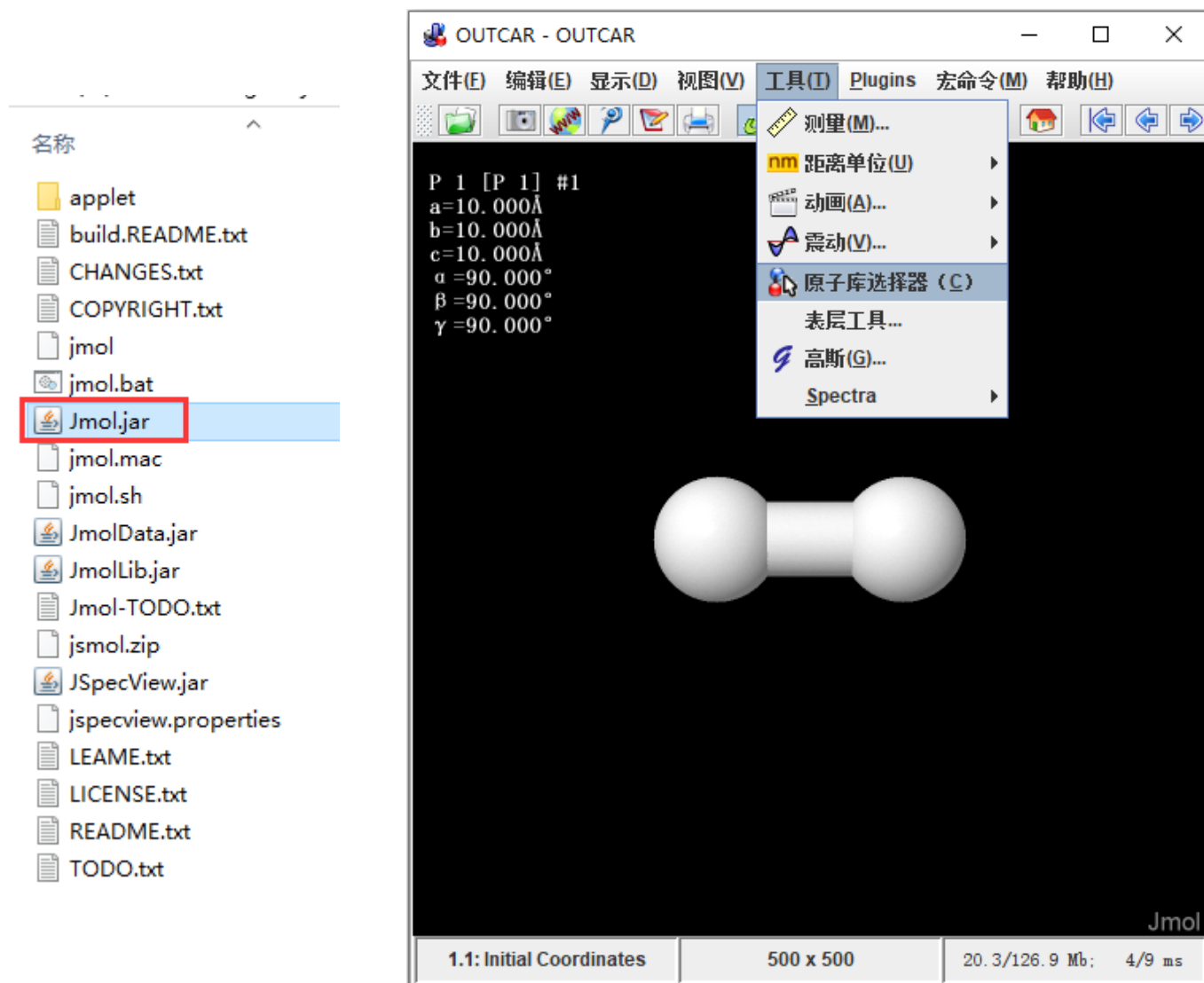
✓ Cite this: *J. Am. Chem. Soc.* 134

RIS Citation GO

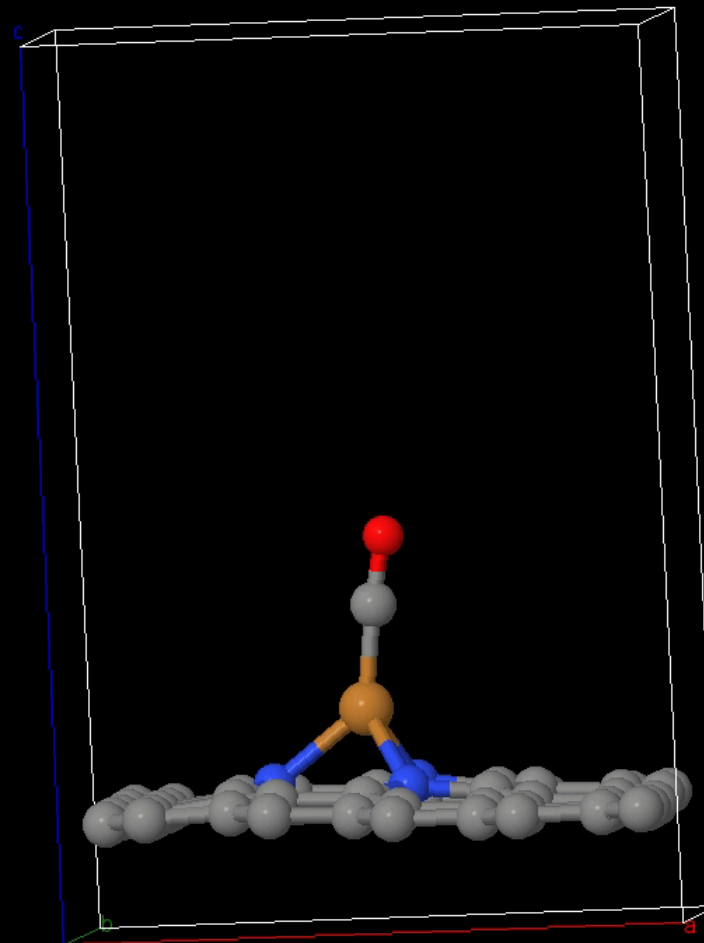
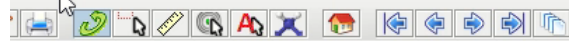
查看振动方式

把OUTCAR直接拖入Jmol里

Jmol可视化软件下载: <https://jmol.sourceforge.net/>

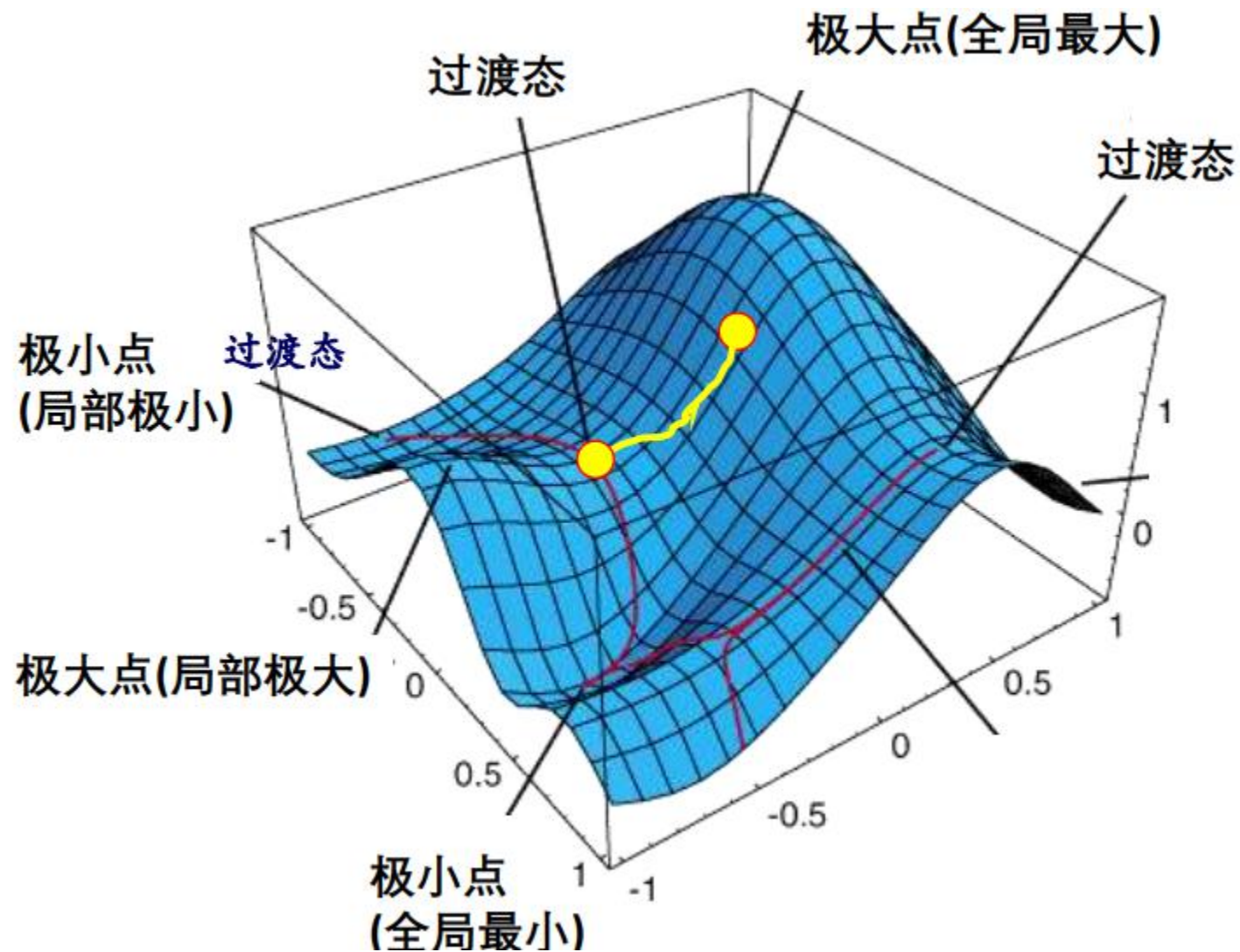


视图(V) 工具(T) Plugins 宏命令(M) 帮助(H)



过渡态搜索

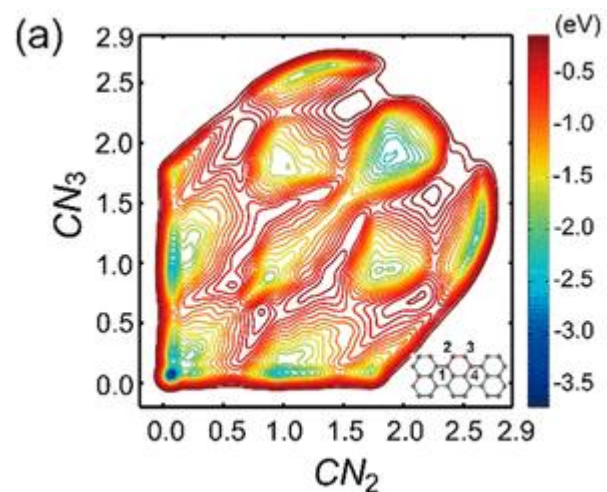
过渡态 (transition state) 搜索，优化到一阶鞍点。过渡态是最小能量路径(MEP)上能量最高的点(一阶鞍点)。在过渡态位置上，仅在沿着反应路径上是能量极大点，而在于之正交的其他所有方向上都是能量极小点



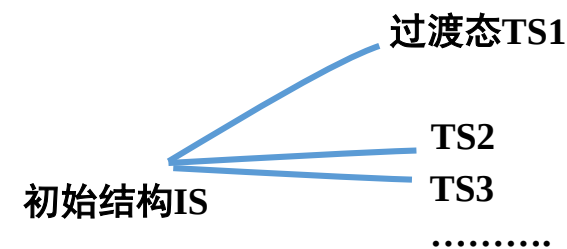
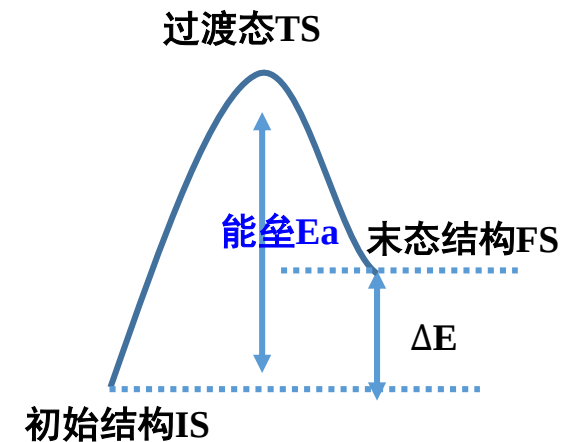
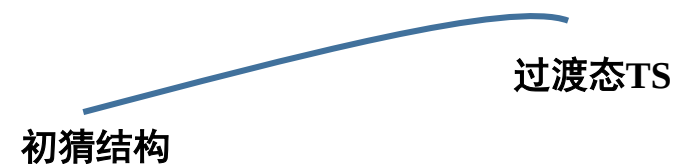
寻找过渡态方法 根据方法所需要的初猜不同，可以分为以下几类：

- (1) 基于单个初猜结构的方法，（DIMER）
- (2) 基于反应物与产物结构的方法，（CI-NEB， DMol3搜索过渡态）
- (3) 基于反应物结构的方法，（SSW）
- (4) 势能面扫描 (meta-dynamics)

过渡态搜索方法集锦：<http://sobereva.com/44>



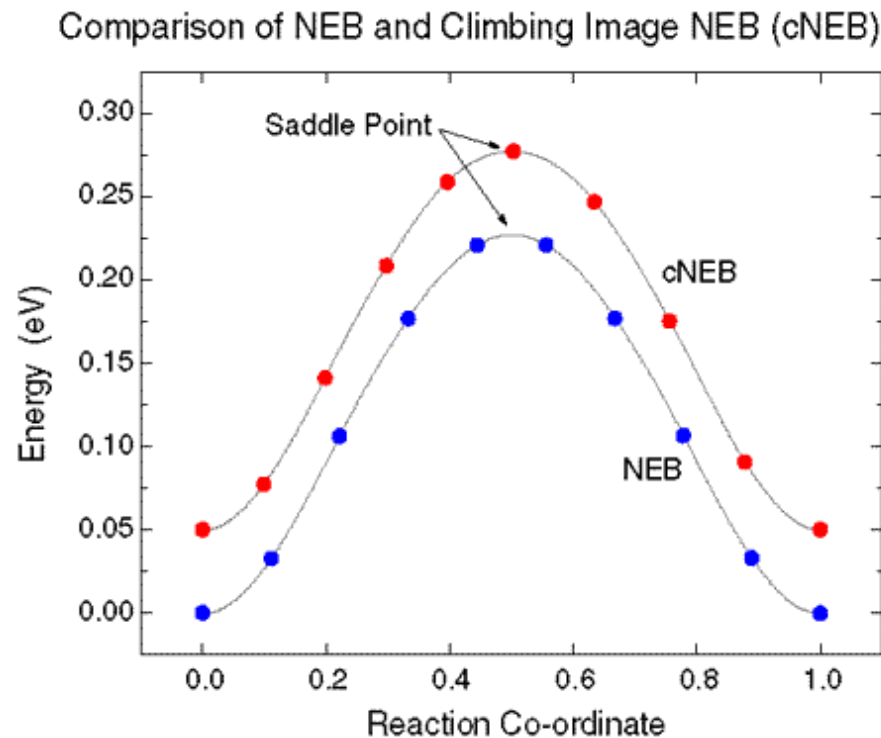
Acc. Chem. Res. 2018, 51, 3, 728–735



NEB (Nudged Elastic Band)

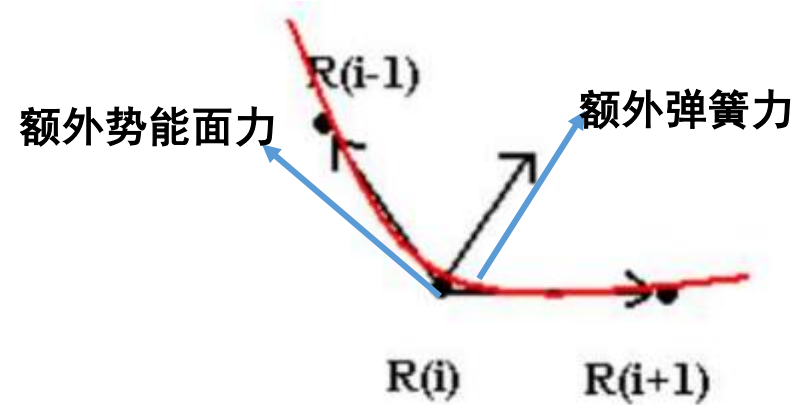
NEB是chain-of-states搜索方法的一种，稳定性好的是CI-NEB方法。

NEB 在初始结构和最终结构之间插入一系列中间结构（称为“副本”或“图像”）共插入 $P-1$ 个，反应物编号为0，产物编号为 P 。它们像一条由弹簧连接的珠子串，形成一条初始路径。接着，通过优化这些中间结构的位置，使整条路径逐渐趋向于最小能量路径。



这P+1点保持**第一和最后一个**点不动，其他的点受到两个力的作用：

1. 来自势能面的力；
2. chain方向上的弹簧力

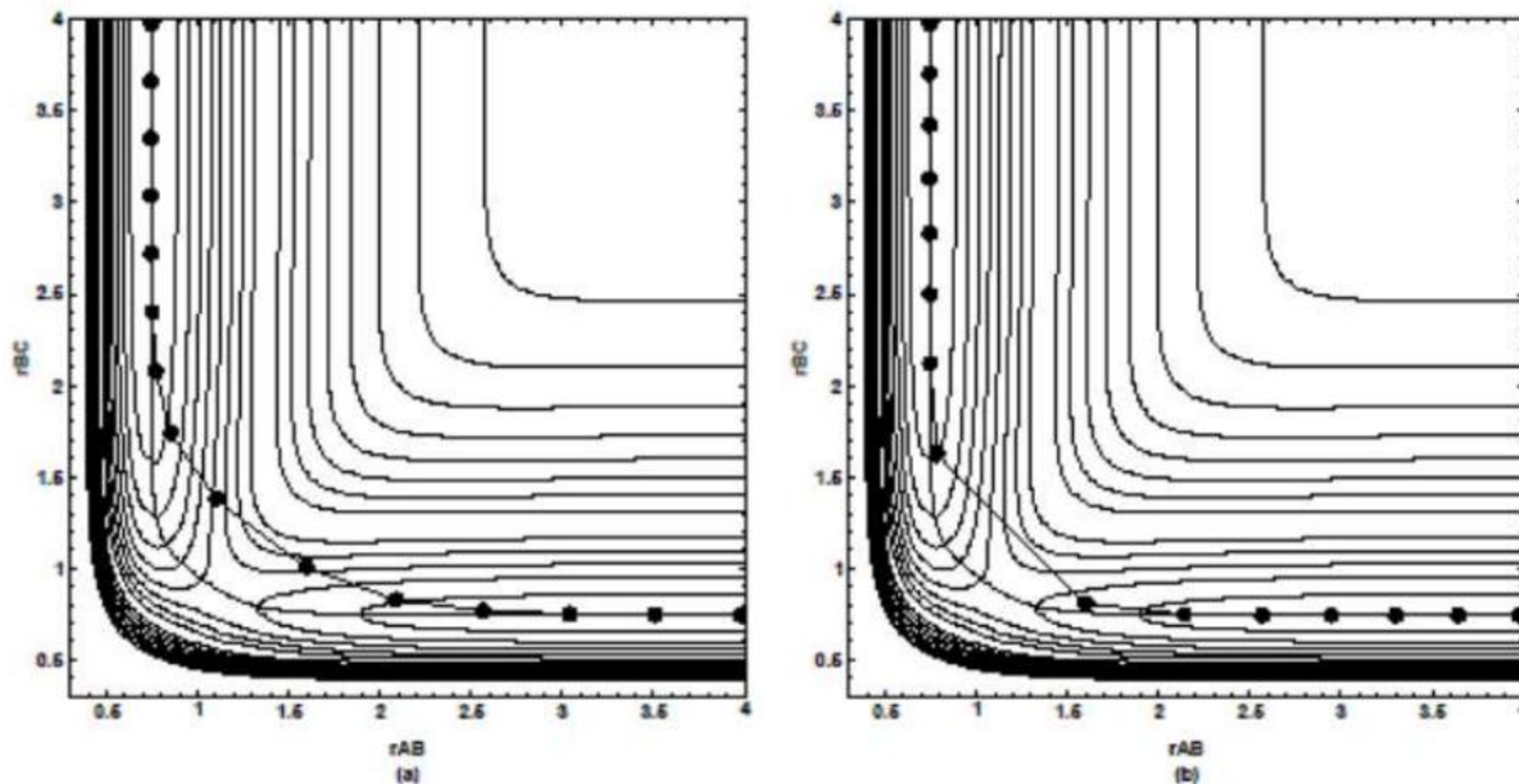


$$S^{PEB}(\vec{R}_1, \dots, \vec{R}_{P-1}) = \sum_{i=0}^P V(\vec{R}_i) + \sum_{i=1}^P \frac{Pk}{2} (\vec{R}_i - \vec{R}_{i-1})^2$$

第一项，相当于让每个珠子都在势能面上向着最稳定的方向上移动。

第二项，含有弹簧力常数，相当于每个点之间的拉力和排斥力，避免珠子之间的距离过短或者过长。

传统PEB方法(plain elastic band)方法遇到的问题：
反应路径偏离minimum energy path(MEP)
难以找到最高点，需要插入非常多的点，浪费计算资源

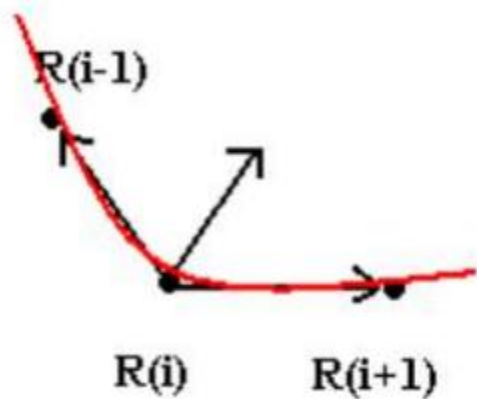


解决方案:

NEB中称为nudge过程

把弹簧力的分量，势能面力的分量投影掉

在势能面拐点的地方会出现两个珠子距离过远的问题，



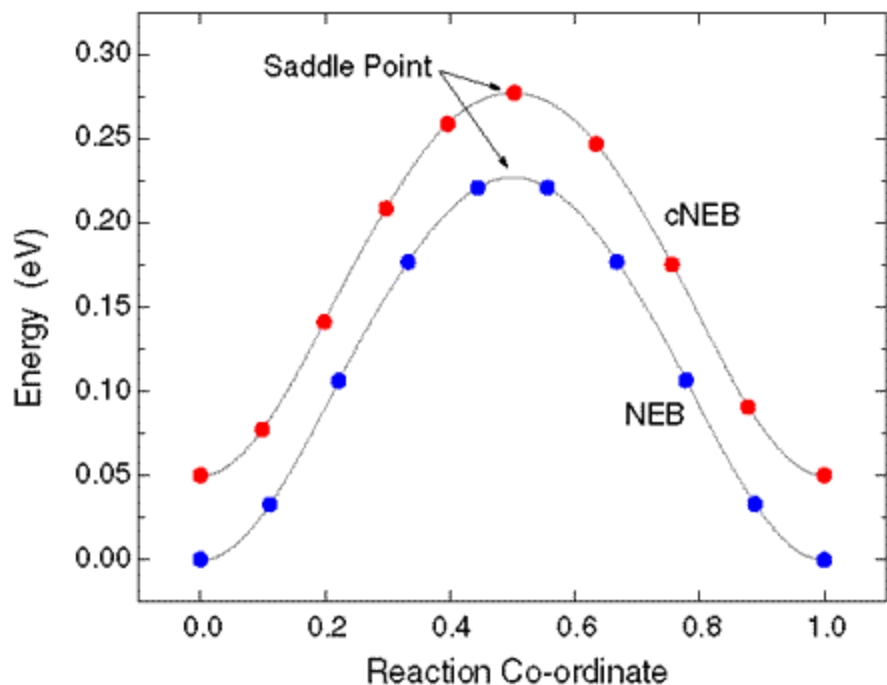
- 弹簧力垂直于路径的分量就被投影掉，
- 弹簧力平行于路径的分量完全保留；
- 势能力在路径方向上的分量被投影掉，
- 势能力垂直于路径上的分量将会引导结构点地正确移动。

以上过程讲述的是VASP原版中使用的NEB。但是此方法仍有很多致命的缺点：

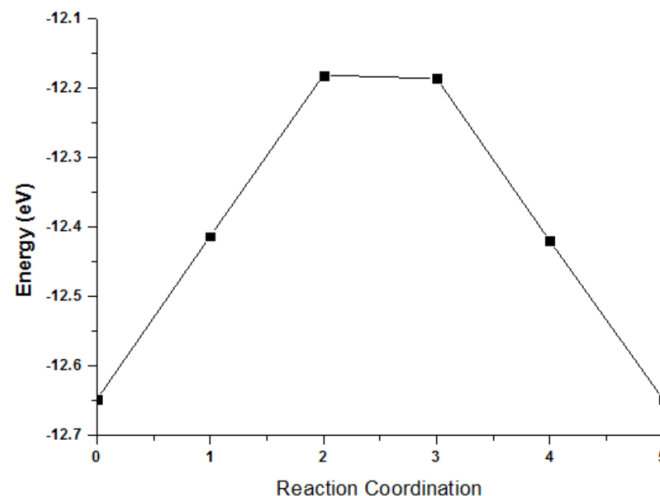
- (1) 要插入足够多的点才可能找到近似的过渡态，计算资源消耗极大。
- (2) 因为珠子不可能到鞍点的顶端，计算出的过渡态能量总是被低估的，又因为本来LDA，GGA泛函就经常低估过渡态能量，所以误差叠加，此方法不太准确。

延伸出CI-NEB (Climbing Image-NEB) 爬坡NEB

Comparison of NEB and Climbing Image NEB (cNEB)



G. Henkelman(VTST的开发者)又发展出了CI-NEB方法，它专门考虑到了定位过渡态问题。CI-NEB与NEB的关键区别是能量最高的点受力的定义，在CI-NEB中这个点不会受到相邻点的弹簧力，避免位置被拉离过渡态，而且将此点平行于路径方向的势能力分量的符号反转，促使此点沿着路径往能量升高的方向上爬到过渡态。



步骤一：优化初态和末态结构。FCC位和HCP位。优化好的结构分别放入00，0x文件夹

步骤二：修改INCAR

传统NEB（VASP自带）

LCLIMB=True. #用CI-NEB方法算TS
ICHAIN=0
IBRION=1
POTIM=0.2 #(0.1~1) 一般0.2可以满足大部分的情况
IMAGES=8 # 反应路径插了8个点
SPRING=-5

IBRION=1（最优路径）

IBRION=3初始和末态结构不是很理想，插的点也是马马虎虎

详细过程参考：

<https://blog.shishiruqi.com/2019/08/19/cineb/>

CI-NEB（VTST）

IOPT=1
ICHAIN=0 #开启NEB方法
LCLIMB=True. ##用CI-NEB方法算TS
IBRION=3 #这是VTST识别并启动VTST优化 算法的标致。
POTIM=0
IMAGES=8 #插点个数
SPRING=-5 #弹簧力常数，用-5默认值即可。
NSW=200
EDIFFG = -0.05 #适当升高力的收敛常数

IOPT = 1，根据建议，选择优化算法：
IOPT设置成1。IOPT推荐7，2，或1。
0意味着启用VASP自带的优化算法，
(比 如： IOPT=0，则IBRION=1, POTIM=0.1)

需确保编译VASP的时候安装了VTST，以下为服务器中vtst版本的vasp

/public3/home/a2s000087/software/vasp.5.4.4_vtst_sol

VTST包含的优化算法：

Optimizer input parameters

The following parameters are read from the INCAR file.

(IOPT = 0) Use VASP optimizers specified from IBRION (default)
(IOPT = 1) LBFGS = Limited-memory Broyden-Fletcher-Goldfarb-Shanno
(IOPT = 2) CG = Conjugate Gradient
(IOPT = 3) QM = Quick-Min
(IOPT = 4) SD = Steepest Descent
(IOPT = 7) FIRE = Fast Inertial Relaxation Engine

常用的是 IOPT = 1, 2 （适合精收敛） IOPT = 7 （适合粗收敛）

preview

- 关掉保存所有的window
- 打开反应物和产物的window，选择window→title vertically
- tool→reaction preview, 选择反应物，产物的文件（display style 选用original模式
- match原子，注意边界原子不要跨越整个supercell
- 匹配完之后，选择需要的images数目，点preview，查看整个路径，看是不是你想要的路径

nebmake.pl POSCAR(初态) POSCAR(末态) N(插点数)

步骤四：初末态对应的CONTCAR, OUTCAR复制到对应的文件夹中，以便后续数据分析。

步骤五：准备POTCAR和KPOINTS 同结构优化直接复制就行

步骤六：提交任务 注意用的核的个数必须整除插点的数量，

步骤七：跟踪检查收敛结果

1) **nebmovie.pl 1** 自动生成一个基于优化好的CONTCAR的movie.xyz文件导入到Jmol可以看 反应路径上所有点的结构

2) **nebef.pl** 可以用nebef.pl或者nebef.s.pl命令（不带参数）查看 计算收敛情况：

0	0.009926	-544.103400	0.000000
1	0.019729	-543.892900	0.210500
2	0.028569	-543.936400	0.167000
3	0.018706	-543.463500	0.639900
4	0.018991	-543.444800	0.658600
5	0.009783	-543.595600	0.507800

后三列： [最大原子受力]、 [能量]、 [相对初态的能量]

当所有插点的最大原子受力都 $< |EDIFFG|$ 时，计算收敛。
若中间插了多个点，则所有点都要 $< |EDIFFG|$ 才能收敛

步骤八：频率计算验证虚频，(有一个振动的虚频)

步骤九：nebresults.pl总结结果 nebresults.pl也是VTST脚本，用来总结neb结果(一般不用，仍用nebef.pl即可)

输出如下：

```
Unzipping the OUTCARs ... done
Do nebbarrier.pl ; nebspline.pl
Do nebef.pl
Do nebmovie.pl
Do nebjmovie.pl
Do nebconverge.pl
Zipping the OUTCARs again ... done

Forces and Energy:
  0      0.000000   -66.750800    0.000000
  1      0.007564   -66.256600    0.494200
  2      0.000000   -66.482400    0.268400

Extremum 1 found at image 0.998685 with energy: 0.494221
Extremum 2 found at image 1.998675 with energy: 0.268416
```


课后作业：

- 完成Cu(111)面吸附CO的计算，判断top, bridge, hollow三种吸附构型哪种更稳定？
- 根据今天课堂的内容，分析最稳定吸附构型的电子结构（电荷密度分布 差分电荷计算）
- **选做：计算吸附体系的吉布斯自由能，分析振动方式**

可参考：<https://www.bilibili.com/video/BV1CN411L7ta>